ENTERPRISE NETWORK DESIGN PROJECT

Secure, Scalable and Highly Available Network for AN Tech

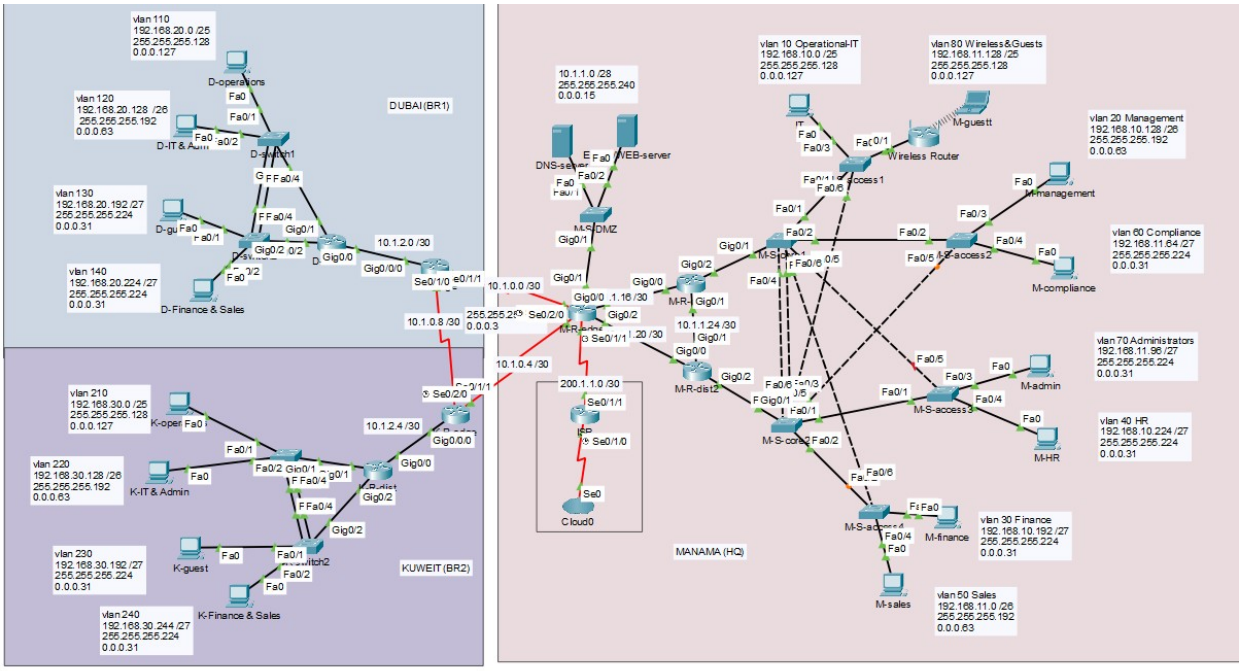
Manama Headquarters • Dubai Branch • Kuwait Branch

Networking and Data Communications

Design, Implementation, Testing and Evaluation Report

Platform: Cisco Packet Tracer

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1. Introduction and Project Scope

1.1 Purpose of this report

This report documents the design, implementation, testing and evaluation of a secure, scalable and highly available enterprise network for **AN Tech**, an organisation with a headquarters in **Manama** and two remote branches in **Dubai** and **Kuwait**. The network was built and tested in **Cisco Packet Tracer**. The report is organised in the same order as the sixteen project requirements. For every requirement it first explains the relevant theory, then compares the available technologies, then justifies the option that was selected, and finally shows how that option is implemented in the running-configuration of the devices.

1.2 Scenario summary

The organisation has several departments, internal services, Internet connectivity and a wireless network at the headquarters. The three sites must communicate securely and reliably, and the design must tolerate growth in staff numbers. The main characteristics of the solution are summarised below.

Table 1: *Solution at a glance*

Aspect	Design choice
Sites	Manama (HQ), Dubai (Branch 1), Kuwait (Branch 2)
Addressing	LAN: `192.168.10.0/24` and `192.168.11.0/24` (HQ), `192.168.20.0/24` (Dubai), `192.168.30.0/24` (Kuwait); infrastructure and DMZ: `10.1.0.0/16`; public WAN link `200.1.1.0/30`
Routing	OSPF process 1, single area (area 0), default route originated by the HQ edge router
Layer 2	VLAN segmentation per department, 802.1Q trunks, Rapid-PVST+ (HQ) and PVST+ (branches), LACP EtherChannel between core/distribution switch pairs
First-hop redundancy	HSRP between `M-R-dist1` and `M-R-dist2` for every HQ VLAN
WAN	Full triangle of serial point-to-point links (HQ–Dubai, HQ–Kuwait, Dubai–Kuwait) giving an alternative path between any two sites
Internet	ISP router at `200.1.1.2/30`; PAT (overload) on `Serial0/1/1` of `M-R-edge`
DMZ	`10.1.1.0/28` (VLAN 15) behind `M-R-edge` hosting the DNS server and the Email/Web server
Security	ACLs, SSH v2 with VTY access-class, sticky port security, banners, secret passwords, shutdown of unused ports, guest/wireless segregation
Wireless	Dedicated VLAN 80 (Wireless & Guests) at HQ, SSID `AN-guests`

1.3 Network topology

Figure 1 shows the logical topology with the subnet allocated to every VLAN and every point-to-point link. **Figure 2** shows the same topology annotated with the physical interface used at each end of every link; it is the reference used throughout this report when a link is described.

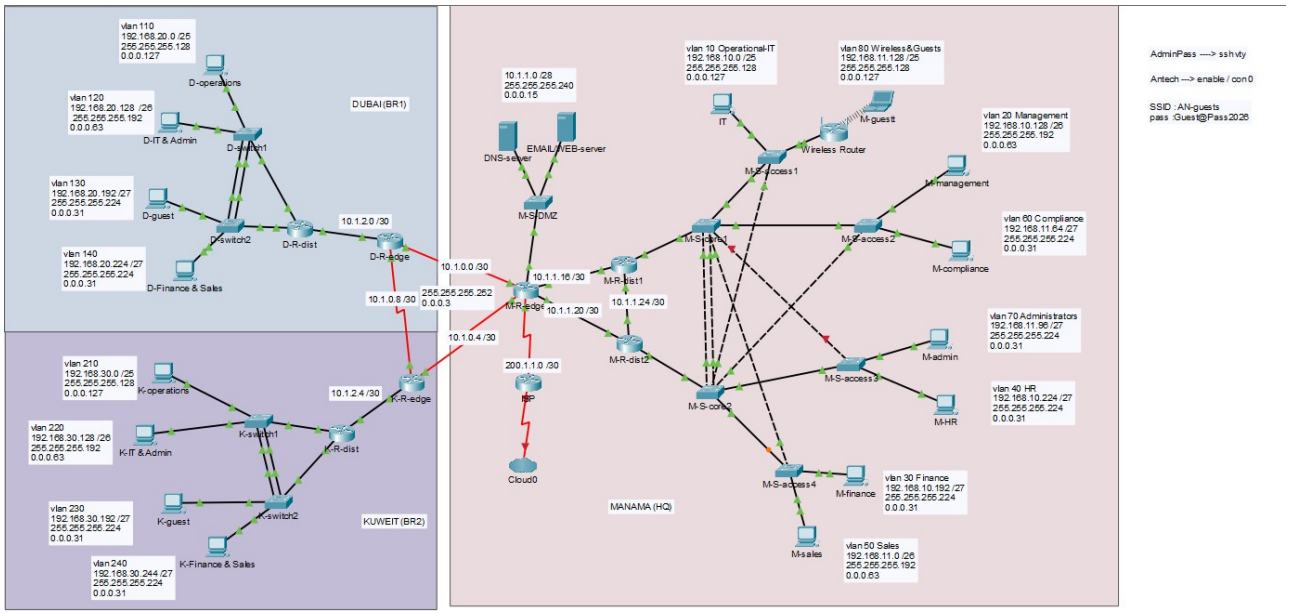


Figure 1: *Logical topology of AN Tech (Manama HQ, Dubai and Kuwait branches, DMZ and ISP)*

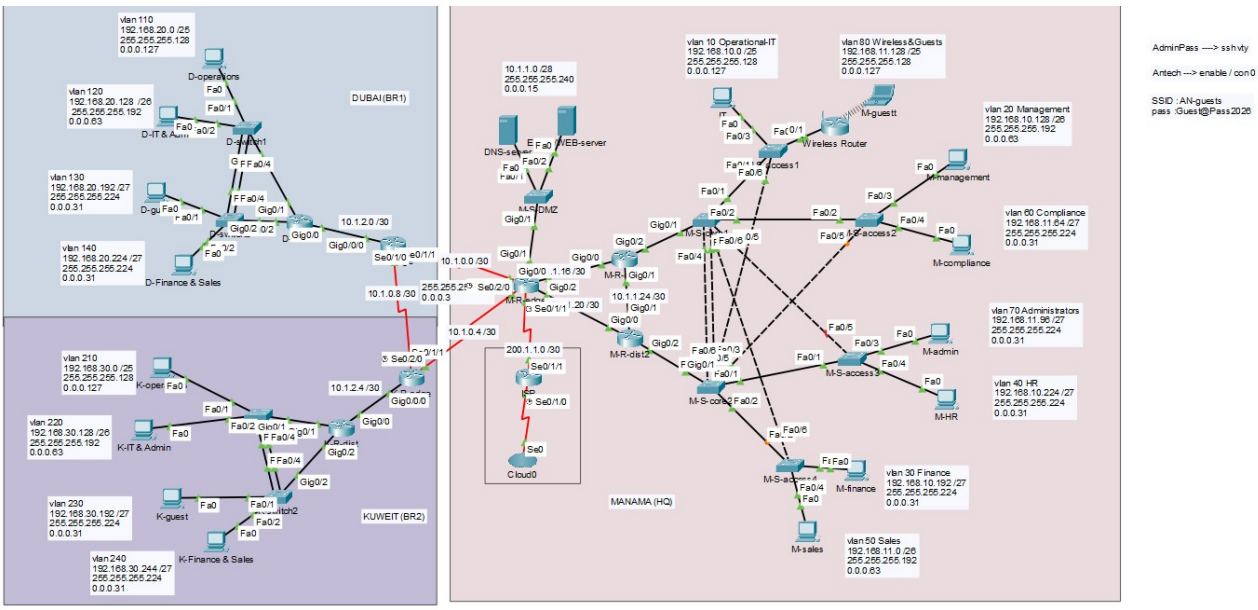


Figure 2: *Physical topology with interface labels*

1.4 Device inventory

The network contains nineteen configured network devices (eight routers and eleven switches) plus end devices (PCs, servers, wireless router and laptop). The table lists every device whose configuration is analysed in this report.

Table 2: *Device inventory*

Device	Model / type	Site	Role
M-R-edge	Cisco 2911 router	Manama	Edge router: WAN hub, NAT/PAT, DMZ gateway, default-route originator
M-R-dist1	Cisco 2911 router	Manama	Distribution router, HSRP primary, inter-VLAN routing, DHCP server
M-R-dist2	Cisco 2911 router	Manama	Distribution router, HSRP standby, inter-VLAN routing
M-S-core1 / M-S-core2	Layer 2 switch	Manama	Core switches (STP root, EtherChannel pair)
M-S-access1 – 4	Layer 2 switch (24 × FE)	Manama	Access switches for departments and wireless router
M-S-DMZ	Layer 2 switch	Manama	DMZ access switch for the DNS and Email/Web servers
ISP	Cisco router (15.4)	Internet	Simulated service provider, `200.1.1.2/30`
D-R-edge	Cisco router (15.4)	Dubai	Branch WAN edge router
D-R-dist	Cisco 2911 router	Dubai	Branch inter-VLAN router (router-on-a-stick), DHCP server
D-switch1 / D-switch2	Layer 2 switch	Dubai	Branch access/aggregation switches
K-R-edge	Cisco router (15.4)	Kuwait	Branch WAN edge router
K-R-dist	Cisco 2911 router	Kuwait	Branch inter-VLAN router (router-on-a-stick), DHCP server
K-switch1 / K-switch2	Layer 2 switch	Kuwait	Branch access/aggregation switches

2. IPv4 Addressing and Routing (Requirement 1)

2.1 Theory: IPv4 addressing, subnetting and VLSM

An IPv4 address is a 32-bit number written as four decimal octets. It is split by the subnet mask into a **network portion** (identifying the subnet) and a **host portion** (identifying the device). Classful addressing (Class A/B/C) wasted space because it forced networks into fixed sizes of /8, /16 and /24. **Classless Inter-Domain Routing (CIDR, RFC 4632)** replaced it: the prefix length can be any value, and contiguous networks can be summarised into one route.

Subnetting divides one address block into smaller subnets so that each broadcast domain has a size that matches its purpose. A subnet with *n* host bits provides 2^n addresses, of which two (the network and broadcast addresses) are not assignable. **Variable Length Subnet Masking (VLSM)** goes one step further: different subnets cut from the same block may have different prefix lengths. A large department receives a /25, a small one a /27, and a router-to-router link receives a /30 (two usable addresses). The result is minimal waste and room to grow.

Table 3: *Subnet sizes used in this design*

Prefix	Mask	Wildcard	Addresses	Usable hosts	Typical use in this project
/25	255.255.255.128	0.0.0.127	128	126	Operational-IT (HQ), Wireless & Guests (HQ), Operations (Dubai, Kuwait)
/26	255.255.255.192	0.0.0.63	64	62	Management, Sales (HQ), IT & Admin (Dubai, Kuwait)
/27	255.255.255.224	0.0.0.31	32	30	Finance, HR, Compliance, Administrators (HQ), Guest and Finance & Sales (branches)
/28	255.255.255.240	0.0.0.15	16	14	DMZ (`10.1.1.0/28`)
/30	255.255.255.252	0.0.0.3	4	2	All router-to-router links, WAN links and the ISP link

Private and public addressing

RFC 1918 reserves three blocks for private use: `10.0.0.0/8`, `172.16.0.0/12` and `192.168.0.0/16`. Private addresses are not routable on the Internet, so they can be reused by any organization, but they must be translated (NAT/PAT, Section 8) at the boundary. Public addresses are globally unique and are allocated by the ISP or a registry. In this project **all internal networks use RFC 1918 space**; the only public-style addressing is the `200.1.1.0/30` link between `M-R-edge` and the ISP, which represents the provider-assigned public address on which PAT is performed.

Table 4: *Use of private address space*

RFC 1918 block	Used for	Reason
`192.168.0.0/16`	All user and guest VLANs at the three sites	Easy-to-read, site-based /24 boundaries: 10.x and 11.x = Manama, 20.x = Dubai, 30.x = Kuwait; the whole supernet is matched by a single wildcard (`0.0.255.255`) in ACLs and NAT
`10.1.0.0/16`	Routers, WAN links, DMZ	Keeps infrastructure and servers in a different range from users, so ACLs and NAT can distinguish them with one statement
Public `200.1.1.0/30`	ISP link	Represents the Internet-facing connection; NAT outside interface

Dynamic routing

Routers learn paths either **statically** (typed by an administrator) or **dynamically** (learned from neighbours using a routing protocol). Dynamic protocols react automatically to link failures, which is essential for the WAN redundancy required in Requirement 4. They are divided into **distance-vector** protocols (RIP, EIGRP) that exchange routing tables with neighbours, and **link-state** protocols (OSPF, IS-IS) in which every router builds the same topology database and runs the Shortest Path First algorithm to compute its routes.

2.2 Addressing plan

The plan was created from the number of hosts each department needs, and then rounded up to the next power of two with spare capacity. Addresses are allocated from the largest subnet to the smallest so that no networks overlap and all blocks stay contiguous, which keeps the OSPF statements and the ACL wildcards short. Gateway addresses follow a fixed convention in every VLAN: the **first usable address is the gateway (HSRP virtual IP at HQ)** and the following addresses are the physical router interfaces.

Table 5: *Site supernets*

Site	User supernet	Infrastructure / transit	Notes
Manama (HQ)	`192.168.10.0/24` + `192.168.11.0/24`	`10.1.1.0/28` (DMZ), `10.1.1.16/30`, `10.1.1.20/30`, `10.1.1.24/30`	Eight user VLANs, largest site
Dubai	`192.168.20.0/24`	`10.1.2.0/30` (edge–dist)	Four VLANs
Kuwait	`192.168.30.0/24`	`10.1.2.4/30` (edge–dist)	Four VLANs
WAN	–	`10.1.0.0/30`, `10.1.0.4/30`, `10.1.0.8/30`	Three inter-site serial links
Internet	–	`200.1.1.0/30`	Link to ISP

Manama (HQ) VLAN subnets

Table 6: *Manama VLAN addressing (derived from the HSRP and DHCP configuration)*

VLAN	Name	Network / prefix	Mask	Usable range	Broadcast	Hosts	Gateway (HSRP VIP)
10	Operational-IT	192.168.10.0/25	255.255.255.128	.1 – .126	192.168.10.127	126	192.168.10.1
20	Management	192.168.10.128/26	255.255.255.192	.129 – .190	192.168.10.191	62	192.168.10.129
30	Finance	192.168.10.192/27	255.255.255.224	.193 – .222	192.168.10.223	30	192.168.10.193
40	HR	192.168.10.224/27	255.255.255.224	.225 – .254	192.168.10.255	30	192.168.10.225
50	Sales	192.168.11.0/26	255.255.255.192	.1 – .62	192.168.11.63	62	192.168.11.1
60	Compliance	192.168.11.64/27	255.255.255.224	.65 – .94	192.168.11.95	30	192.168.11.65
70	Administrators	192.168.11.96/27	255.255.255.224	.97 – .126	192.168.11.127	30	192.168.11.97
80	Wireless & Guests	192.168.11.128/25	255.255.255.128	.129 – .254	192.168.11.255	126	192.168.11.129
15	DMZ	10.1.1.0/28	255.255.255.240	.1 – .14	10.1.1.15	14	10.1.1.1 (M-R-edge)

Dubai and Kuwait VLAN subnets

Table 7: *Branch VLAN addressing*

Site	VLAN	Name	Network / prefix	Usable range	Broadcast	Hosts	Gateway
Dubai	110	Operations	192.168.20.0/25	.1 – .126	192.168.20.127	126	192.168.20.1
Dubai	120	IT & Admin	192.168.20.128/26	.129 – .190	192.168.20.191	62	192.168.20.129
Dubai	130	Guest	192.168.20.192/27	.193 – .222	192.168.20.223	30	192.168.20.193

Site	VLAN	Name	Network / prefix	Usable range	Broadcast	Hosts	Gateway
Dubai	140	Finance & Sales	192.168.20.224/27	.225 – .254	192.168.20.255	30	192.168.20.225
Kuwait	210	Operations	192.168.30.0/25	.1 – .126	192.168.30.127	126	192.168.30.1
Kuwait	220	IT & Admin	192.168.30.128/26	.129 – .190	192.168.30.191	62	192.168.30.129
Kuwait	230	Guest	192.168.30.192/27	.193 – .222	192.168.30.223	30	192.168.30.193
Kuwait	240	Finance & Sales	192.168.30.224/27	.225 – .254	192.168.30.255	30	192.168.30.225

Support for current users and future growth

Each subnet is deliberately larger than the number of hosts presently connected, and every site keeps neighbouring address space unused. The DHCP configuration in Section 14 also reserves the first addresses of every VLAN for gateways, switches and static devices, so growth in user numbers does not require renumbering.

Table 8: *Growth strategy*

Site	Currently allocated	Unused adjacent space available for expansion	Expansion approach
Manama	`192.168.10.0` – `192.168.11.255`	`192.168.12.0/24` upwards (up to `192.168.19.255`)	New department = new VLAN and /27–/26 from the free range; or widen a VLAN into a larger prefix before it is exhausted
Dubai	`192.168.20.0/24` (fully carved)	`192.168.21.0/24`	Add a /24 and advertise it with one OSPF `network` statement
Kuwait	`192.168.30.0/24` (fully carved)	`192.168.31.0/24`	Same method as Dubai
Transit links	`10.1.0.0/30`, `.4/30`, `.8/30`; `10.1.1.x`; `10.1.2.x`	Rest of `10.1.0.0/16`	Additional routers or links use further /30 blocks

Because every site uses a distinct third octet range, a future summary route can be advertised per site (for example `192.168.20.0/24` for Dubai), which reduces the size of routing tables if more branches are added.

2.3 Link and interface addressing

Every router-to-router link uses a /30, which provides exactly two usable addresses and wastes nothing. The table lists all point-to-point links with the interface and address at each end, taken from the running-configurations.

Table 9: *Point-to-point and transit links*

Link	Subnet	End A (interface / IP)	End B (interface / IP)	Purpose
HQ – Dubai	10.1.0.0/30	M-R-edge `Se0/1/0` 10.1.0.1	D-R-edge `Se0/1/0` 10.1.0.2	WAN (serial)
HQ – Kuwait	10.1.0.4/30	M-R-edge `Se0/2/0` 10.1.0.5	K-R-edge `Se0/2/0` 10.1.0.6	WAN (serial)
Dubai – Kuwait	10.1.0.8/30	D-R-edge `Se0/1/1` 10.1.0.9	K-R-edge `Se0/1/1` 10.1.0.10	WAN (serial), alternative path
HQ edge – dist1	10.1.1.16/30	M-R-edge `Gi0/0` 10.1.1.17	M-R-dist1 `Gi0/0` 10.1.1.18	LAN uplink
HQ edge – dist2	10.1.1.20/30	M-R-edge `Gi0/2` 10.1.1.21	M-R-dist2 `Gi0/0` 10.1.1.22	LAN uplink
HQ dist1 – dist2	10.1.1.24/30	M-R-dist1 `Gi0/1` 10.1.1.25	M-R-dist2 `Gi0/1` 10.1.1.26	Distribution inter-link
Dubai edge – dist	10.1.2.0/30	D-R-edge `Gi0/0/0` 10.1.2.1	D-R-dist `Gi0/0` 10.1.2.2	Branch LAN uplink
Kuwait edge – dist	10.1.2.4/30	K-R-edge `Gi0/0/0` 10.1.2.5	K-R-dist `Gi0/0` 10.1.2.6	Branch LAN uplink
HQ – ISP	200.1.1.0/30	M-R-edge `Se0/1/1` 200.1.1.1	ISP `Se0/1/1` 200.1.1.2	Internet access (NAT outside)

Link	Subnet	End A (interface / IP)	End B (interface / IP)	Purpose
HQ DMZ	10.1.1.0/28	M-R-edge `Gi0/1.15` 10.1.1.1	DNS 10.1.1.2; Email/Web 10.1.1.3	DMZ servers (VLAN 15)

The complete per-interface listing for every router is given in Appendix A, and the management addresses of the switches are given in Appendix B.

2.4 Inter-VLAN routing

Hosts in different VLANs cannot communicate at Layer 2, so a Layer 3 device must route between them. Three methods exist: (1) one physical router interface per VLAN, (2) **router-on-a-stick** – a single trunk link to a router that has one 802.1Q sub-interface per VLAN, and (3) a multilayer switch with Switched Virtual Interfaces (SVIs).

Table 10: *Inter-VLAN routing options*

Method	Advantages	Disadvantages	Decision
One router port per VLAN	Simple, no tagging	Needs one port per VLAN; impossible with 8 VLANs on a 2911	Rejected
Router-on-a-stick	One link, works with the available Layer 2 switches, easy to read and troubleshoot, ACLs applied per sub-interface	The trunk is shared bandwidth and a single point of failure per router	Selected (mitigated at HQ by two routers + HSRP)
Multilayer switch SVIs	Hardware-speed routing, best scalability	Requires Layer 3 switches, which are not part of the device set used	Recommended future upgrade

Router-on-a-stick was selected because the topology uses 2911 routers and Layer 2 switches. At HQ the routing is performed by **two** routers (`M-R-dist1` and `M-R-dist2`) which both have sub-interfaces for every VLAN, so the loss of one router does not stop inter-VLAN routing. At the branches a single router (`D-R-dist`, `K-R-dist`) performs the function.

Table 11: *Router-on-a-stick sub-interfaces*

Router	Physical interface	Sub-interfaces (dot1Q VLAN → IP/mask)
M-R-dist1	Gi0/2	.10 → 192.168.10.2/25; .20 → 192.168.10.130/26; .30 → 192.168.10.194/27; .40 → 192.168.10.226/27; .50 → 192.168.11.2/26; .60 → 192.168.11.66/27; .70 → 192.168.11.98/27; .80 → 192.168.11.130/25
M-R-dist2	Gi0/2	.10 → 192.168.10.3/25; .20 → 192.168.10.131/26; .30 → 192.168.10.195/27; .40 → 192.168.10.227/27; .50 → 192.168.11.3/26; .60 → 192.168.11.67/27; .70 → 192.168.11.99/27; .80 → 192.168.11.131/25
M-R-edge	Gi0/1	.15 → 10.1.1.1/28 (DMZ); .70 → 192.168.11.97/27 (see action note above)
D-R-dist	Gi0/1	.110 → 192.168.20.1/25; .120 → 192.168.20.129/26; .130 → 192.168.20.193/27; .140 → 192.168.20.225/27
K-R-dist	Gi0/1	.210 → 192.168.30.1/25; .220 → 192.168.30.129/26; .230 → 192.168.30.193/27; .240 → 192.168.30.225/27

```
interface GigabitEthernet0/2.30
 encapsulation dot1Q 30
 ip address 192.168.10.194 255.255.255.224
 standby 30 ip 192.168.10.193
 standby 30 priority 110
 standby 30 preempt
```

The excerpt above (from `M-R-dist1`) shows the pattern used for all sub-interfaces: the VLAN is bound with `encapsulation dot1Q`, a unique address is given to the router, and the shared gateway is the HSRP virtual address (Section 12).

2.5 Selection of the dynamic routing protocol

OSPF (RFC 2328) was selected for the following reasons.

- **Fast convergence and automatic failover:** the WAN is a triangle of three serial links; if one fails OSPF recalculates and traffic uses the other two links without any manual change (Requirement 4 and 6).
- **Bandwidth-aware metric:** cost is derived from interface bandwidth, so Gigabit paths are preferred over serial paths, and a direct serial link (cost 64) is preferred over a two-hop path (cost 128).
- **Open standard:** unlike EIGRP it works with routers from any vendor, which protects the organisation if different equipment is purchased for a future branch.
- **Scalable:** all sites currently fit comfortably in a single area; the OSPF area design allows additional areas later (for example one per branch) without changing the protocol.
- **Support for default-route origination:** `M-R-edge` injects a default route so that the branches and HQ distribution routers learn the path to the Internet dynamically.

OSPF implementation

All routers run OSPF process 1 in area 0. A single area was chosen because the network has fewer than ten routers, which keeps the link-state database small and the configuration easy to verify. The `network` statements use exact wildcard masks, so that only the intended interfaces take part in OSPF and no unexpected network is advertised.

Table 13: *OSPF configuration per router (from the running-config)*

Router	Networks advertised (area 0)	Other OSPF settings
M-R-edge	10.1.0.0/30, 10.1.0.4/30, 10.1.1.16/30, 10.1.1.20/30, 10.1.1.0/28	`default-information originate`; static `ip route 0.0.0.0 0.0.0.0 200.1.1.2`
M-R-dist1	All eight HQ VLAN subnets, 10.1.1.16/30, 10.1.1.24/30	`passive-interface default`; `no passive-interface` on `Gi0/0` and `Gi0/1`
M-R-dist2	All eight HQ VLAN subnets, 10.1.1.20/30, 10.1.1.24/30	`passive-interface default`; `no passive-interface` on `Gi0/0` and `Gi0/1`
D-R-edge	10.1.0.0/30, 10.1.0.8/30, 10.1.2.0/30	–
D-R-dist	192.168.20.0/25, .128/26, .192/27, .224/27, 10.1.2.0/30 (plus 10.1.0.0/30 and 10.1.0.8/30)	–
K-R-edge	10.1.0.4/30, 10.1.0.8/30, 10.1.2.4/30	–
K-R-dist	192.168.30.0/25, .128/26, .192/27, .224/27, 10.1.2.4/30	–

```
router ospf 1
 log-adjacency-changes
 passive-interface default
 no passive-interface GigabitEthernet0/0
 no passive-interface GigabitEthernet0/1
 network 192.168.10.0 0.0.0.127 area 0
 network 192.168.10.128 0.0.0.63 area 0
 network 192.168.10.192 0.0.0.31 area 0
 network 192.168.10.224 0.0.0.31 area 0
 network 192.168.11.0 0.0.0.63 area 0
 network 192.168.11.64 0.0.0.31 area 0
 network 192.168.11.96 0.0.0.31 area 0
 network 192.168.11.128 0.0.0.127 area 0
 network 10.1.1.16 0.0.0.3 area 0
 network 10.1.1.24 0.0.0.3 area 0
```

The excerpt (from `M-R-dist1`) shows two good practices: **passive-interface default** stops OSPF hello packets being sent into user VLANs (reducing overhead and preventing a rogue host from forming an adjacency), and only the two uplinks that need neighbours are re-enabled.

Table 14: *Path selection and failover*

Traffic	Preferred path (cost)	Path if the preferred WAN link fails
Manama ↔ Dubai	Direct serial 10.1.0.0/30 (64)	Manama → Kuwait → Dubai over 10.1.0.4/30 and 10.1.0.8/30 (128)
Manama ↔ Kuwait	Direct serial 10.1.0.4/30 (64)	Manama → Dubai → Kuwait (128)
Dubai ↔ Kuwait	Direct serial 10.1.0.8/30 (64)	Dubai → Manama → Kuwait (128)
Any site → Internet	Via `M-R-edge` using the OSPF default route	No alternative: `M-R-edge` is the only Internet gateway (see Section 5)

2.6 Connectivity verification

Connectivity is verified in layers: first the local gateway, then another VLAN at the same site, then the other sites, then the DMZ and the Internet. The expected result for each test is shown below; the full test log with pass/fail status is in Section 16.

Table 15: *Addressing and routing tests*

#	Test	Command / action	Expected result
1.1	Host reaches default gateway	PC: `ping <HSRP VIP>`	Reply
1.2	Inter-VLAN at HQ (e.g. Sales → Compliance)	PC: `ping 192.168.11.x`	Reply
1.3	Inter-site (HQ → Dubai / Kuwait)	PC: `ping 192.168.20.x`, `ping 192.168.30.x`	Reply
1.4	OSPF neighbours	`show ip ospf neighbor` on all routers	State FULL with every directly connected router
1.5	Routing table	`show ip route ospf`	All site subnets present as `O`; default `O*E2` on dist and branch routers
1.6	Path trace between sites	`tracert 192.168.30.x` from Dubai	Direct path via 10.1.0.8/30

```
M-R-edge#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.11.130    1   FULL/BDR        00:00:38    10.1.1.18
GigabitEthernet0/0
10.1.2.1          0   FULL/  -        00:00:38    10.1.0.2        Serial0/1/0
10.1.2.5          0   FULL/  -        00:00:38    10.1.0.6        Serial0/2/0
192.168.11.131    1   FULL/BDR        00:00:38    10.1.1.22
GigabitEthernet0/2
```

```
M-R-edge#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 200.1.1.2 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 14 subnets, 3 masks
C       10.1.0.0/30 is directly connected, Serial0/1/0
L       10.1.0.1/32 is directly connected, Serial0/1/0
C       10.1.0.4/30 is directly connected, Serial0/2/0
L       10.1.0.5/32 is directly connected, Serial0/2/0
O       10.1.0.8/30 [110/128] via 10.1.0.2, 00:23:53, Serial0/1/0
                    [110/128] via 10.1.0.6, 00:23:53, Serial0/2/0
C       10.1.1.0/28 is directly connected, GigabitEthernet0/1.15
L       10.1.1.1/32 is directly connected, GigabitEthernet0/1.15
C       10.1.1.16/30 is directly connected, GigabitEthernet0/0
L       10.1.1.17/32 is directly connected, GigabitEthernet0/0
C       10.1.1.20/30 is directly connected, GigabitEthernet0/2
L       10.1.1.21/32 is directly connected, GigabitEthernet0/2
O       10.1.1.24/30 [110/2] via 10.1.1.18, 00:23:10, GigabitEthernet0/0
                     [110/2] via 10.1.1.22, 00:23:10, GigabitEthernet0/2
O       10.1.2.0/30 [110/65] via 10.1.0.2, 00:23:10, Serial0/1/0
O       10.1.2.4/30 [110/65] via 10.1.0.6, 00:23:10, Serial0/2/0
     192.168.10.0/24 is variably subnetted, 4 subnets, 3 masks
O       192.168.10.0/25 [110/2] via 10.1.1.18, 00:23:10, GigabitEthernet0/0
                        [110/2] via 10.1.1.22, 00:23:10, GigabitEthernet0/2
O       192.168.10.128/26 [110/2] via 10.1.1.18, 00:23:10, GigabitEthernet0/0
                          [110/2] via 10.1.1.22, 00:23:10, GigabitEthernet0/2
O       192.168.10.192/27 [110/2] via 10.1.1.18, 00:23:10, GigabitEthernet0/0
                          [110/2] via 10.1.1.22, 00:23:10, GigabitEthernet0/2
O       192.168.10.224/27 [110/2] via 10.1.1.18, 00:23:10, GigabitEthernet0/0
                          [110/2] via 10.1.1.22, 00:23:10, GigabitEthernet0/2
     192.168.11.0/24 is variably subnetted, 5 subnets, 4 masks
O       192.168.11.0/26 [110/2] via 10.1.1.18, 00:23:10, GigabitEthernet0/0
                        [110/2] via 10.1.1.22, 00:23:10, GigabitEthernet0/2
O       192.168.11.64/27 [110/2] via 10.1.1.18, 00:23:10, GigabitEthernet0/0
                         [110/2] via 10.1.1.22, 00:23:10, GigabitEthernet0/2
C       192.168.11.96/27 is directly connected, GigabitEthernet0/1.70
L       192.168.11.97/32 is directly connected, GigabitEthernet0/1.70
O       192.168.11.128/25 [110/2] via 10.1.1.18, 00:23:10, GigabitEthernet0/0
                          [110/2] via 10.1.1.22, 00:23:10, GigabitEthernet0/2
     192.168.20.0/24 is variably subnetted, 4 subnets, 3 masks
O       192.168.20.0/25 [110/66] via 10.1.0.2, 00:23:10, Serial0/1/0
O       192.168.20.128/26 [110/66] via 10.1.0.2, 00:23:10, Serial0/1/0
O       192.168.20.192/27 [110/66] via 10.1.0.2, 00:23:10, Serial0/1/0
O       192.168.20.224/27 [110/66] via 10.1.0.2, 00:23:10, Serial0/1/0
     192.168.30.0/24 is variably subnetted, 4 subnets, 3 masks
O       192.168.30.0/25 [110/66] via 10.1.0.6, 00:23:10, Serial0/2/0
O       192.168.30.128/26 [110/66] via 10.1.0.6, 00:23:10, Serial0/2/0
O       192.168.30.192/27 [110/66] via 10.1.0.6, 00:23:10, Serial0/2/0
O       192.168.30.224/27 [110/66] via 10.1.0.6, 00:23:10, Serial0/2/0
     200.1.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.1.1.0/30 is directly connected, Serial0/1/1
L       200.1.1.1/32 is directly connected, Serial0/1/1
S*   0.0.0.0/0 [1/0] via 200.1.1.2

M-R-edge#
```

Figure 3: *Evidence – `show ip ospf neighbor` and `show ip route` on M-R-edge*

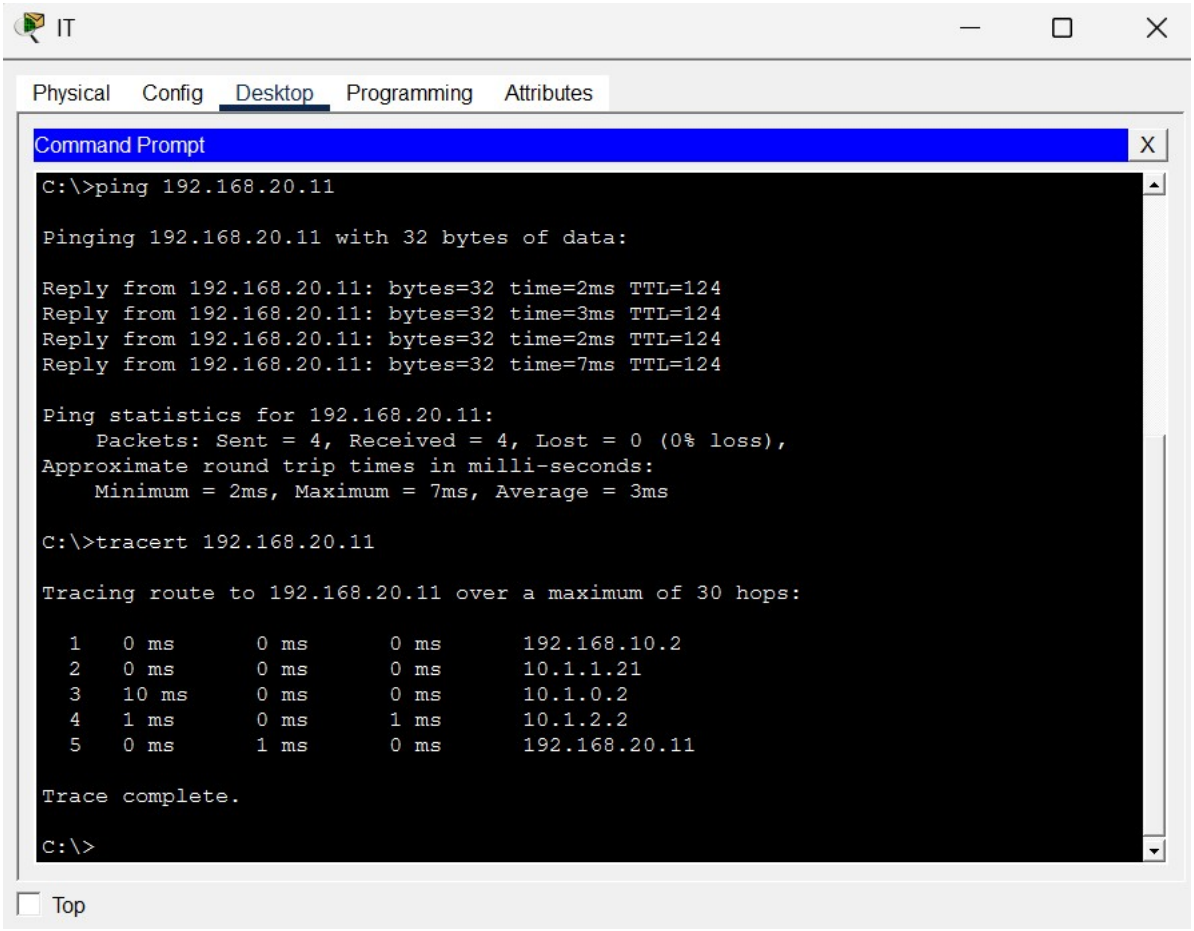


Figure 4: *Evidence – Ping and tracert between a Manama host and a Dubai host*

3. Network Optimization and VLAN Design (Requirement 2)

3.1 Theory: broadcast domains, VLANs, trunks, native VLAN, VTP and STP

A **broadcast domain** is the set of devices that receive a Layer 2 broadcast frame sent by any member. Every ARP request, DHCP discovery and many service announcements are broadcasts, so in a large flat network every host is interrupted by traffic that does not concern it, and a faulty NIC or loop can flood the entire network. Routers do not forward broadcasts, so each router interface is a separate broadcast domain.

A **VLAN (IEEE 802.1Q)** creates several logical broadcast domains on the same physical switches. A switch port can be an **access port** (belongs to one VLAN, carries untagged frames for an end device) or a **trunk port** (carries frames of many VLANs between switches or to a router, each tagged with a 4-byte 802.1Q header). The **native VLAN** is the one VLAN whose frames cross a trunk untagged; if it is left as the default VLAN 1, it can be abused for VLAN-hopping attacks, so a dedicated unused VLAN should be used on both ends of every trunk.

VTP (VLAN Trunking Protocol) distributes the VLAN database between switches. In **server** and **client** mode, a switch with a higher revision number overwrites the VLAN database of every other switch in the domain – a single mistaken change can delete VLANs network-wide. **Transparent** mode does not take part in synchronisation (it still forwards VTP advertisements), so each switch is managed independently. **Off** mode disables VTP entirely.

Table 16: *VTP modes compared*

Mode	Creates / edits VLANs	Synchronises with others	Risk	Suitability here
Server	Yes	Yes (advertises)	High – accidental overwrite	No
Client	No	Yes (accepts)	High – can be overwritten by a rogue server	No
Transparent	Yes (local only)	No	Low	**Selected**: small number of switches per site, full control
Off	Yes (local only)	No	Lowest	Also acceptable

Spanning Tree Protocol (STP, IEEE 802.1D) prevents Layer 2 loops in networks with redundant links by electing a **root bridge** and blocking redundant ports. Cisco implements it per VLAN (PVST+). **Rapid PVST+ (IEEE 802.1w)** converges in about one to two seconds instead of 30–50 seconds.

Table 17: *STP variants compared*

Variant	Convergence	Instances	Notes
802.1D (CST)	30–50 s	One for all VLANs	Wastes redundant links, slow
PVST+	30–50 s	One per VLAN	Allows different root per VLAN (load sharing); used in Dubai/Kuwait/DMZ
Rapid PVST+	1–2 s	One per VLAN	Fast, backward compatible with PVST+; used at HQ
MSTP (802.1s)	Fast	Groups of VLANs	Best for very many VLANs; unnecessary complexity here

3.2 VLAN design and segmentation

Users are separated by **department and trust level**. Operational and administrative users, finance, HR and compliance staff are in different VLANs from each other and from guests, so that policy (ACLs, port security,

monitoring) can be applied per VLAN and a broadcast storm in one department cannot affect the others. Guests and wireless users receive their own VLAN at every site. The same pattern (operations / IT & admin / guest / finance & sales) is repeated in the branches so that administrators work with the same logic at all three locations.

Table 18: *VLAN plan*

Site	VLAN	Name	Subnet	Trust level	Purpose
Manama	10	Operational-IT	192.168.10.0/25	Internal	IT operations staff
Manama	20	Management	192.168.10.128/26	Internal	Management staff
Manama	30	Finance	192.168.10.192/27	Sensitive	Finance department
Manama	40	HR	192.168.10.224/27	Sensitive	Human Resources
Manama	50	Sales	192.168.11.0/26	Internal	Sales department
Manama	60	Compliance	192.168.11.64/27	Sensitive	Compliance department
Manama	70	Administrators	192.168.11.96/27	Management plane	Network administrators; the only VLAN allowed to SSH to devices
Manama	80	Wireless & Guests	192.168.11.128/25	Untrusted	Wi-Fi users and guests
Manama	15	DMZ	10.1.1.0/28	Semi-trusted	Public-facing servers
Dubai	110 / 120 / 130 / 140	Operations / IT & Admin / Guest / Finance & Sales	192.168.20.0/24 split	130 untrusted	Branch departments
Kuwait	210 / 220 / 230 / 240	Operations / IT & Admin / Guest / Finance & Sales	192.168.30.0/24 split	230 untrusted	Branch departments
All	58 (HQ) / 99 (branches)	Unused-port parking VLAN	–	Isolated	All unused access ports are placed here and shut down

How segmentation improves performance, security and manageability

Table 19: *Benefits of segmentation*

Benefit	Explanation in this network
Performance	The largest broadcast domain is a /25 (126 hosts) instead of one flat network of every host at the site. ARP/DHCP broadcasts from one department are not delivered to other departments, and a loop or faulty NIC is contained in one VLAN.
Security	Traffic between departments must pass a router, where ACLs can filter it. Finance, HR and Compliance are separate from guests and wireless users. Unused ports are isolated in a parking VLAN and shut down.
Manageability	Each department is one subnet with a predictable gateway, a dedicated DHCP pool and its own HSRP group. Changes (new users, new policy) are made on one VLAN without touching others.
Troubleshooting	A fault is localised quickly: the VLAN, subnet and gateway identify the department immediately.

3.3 Access and trunk port configuration

Host-facing ports are configured as access ports in one VLAN. Links between switches, and links from a switch to a router, are configured as trunks that carry **only the VLANs that are needed on that link** (`switchport trunk allowed vlan`). Pruning the allowed list limits unnecessary broadcast propagation and prevents unwanted VLANs from reaching a switch. For example, `M-S-access1` serves VLANs 10 and 80 only, so its uplinks allow only those two VLANs.

Table 20: *Trunk ports (from the running-config)*

Switch	Trunk interface(s)	Connects to	Allowed VLANs
M-S-core1	Gi0/1	M-R-dist1 `Gi0/2`	10,20,30,40,50,60,70,80

Switch	Trunk interface(s)	Connects to	Allowed VLANs
M-S-core1	Po1 (Fa0/3, Fa0/4)	M-S-core2 (EtherChannel)	10,20,30,40,50,60,70,80
M-S-access1	Fa0/1, Fa0/6	Core switches (one uplink each)	10, 80
M-S-access2	Fa0/2, Fa0/5	Core switches (one uplink each)	20, 60
M-S-access3	Fa0/1	Core switch	40, 70
M-S-access4	Fa0/2, Fa0/6	Core switches (one uplink each)	30, 50
M-S-DMZ	Gi0/1	M-R-edge `Gi0/1`	15, 70
D-switch1	Gi0/1; Po1 (Fa0/3, Fa0/4)	D-R-dist; D-switch2	110,120,130,140
D-switch2	Po1 (Fa0/3, Fa0/4)	D-switch1	All (no restriction)
K-switch1	Gi0/1; Po1 (Fa0/3, Fa0/4)	K-R-dist; K-switch2	All (no restriction)
K-switch2	Gi0/2; Po1 (Fa0/3, Fa0/4)	K-R-dist; K-switch1	All (no restriction)

Table 21: *Access ports with hosts (from the running-config and topology)*

Switch	Port	VLAN	Connected device	Sticky MAC learned
M-S-access1	Fa0/3	10	M-IT PC	0003.E49C.D7C6
M-S-access1	Fa0/4	80	Wireless router	Sticky enabled, none recorded
M-S-access2	Fa0/3	20	M-management PC	None recorded at capture
M-S-access2	Fa0/4	60	M-compliance PC	0005.5E45.7022
M-S-access3	Fa0/3	70	M-admin PC	0002.4AE6.C3CC
M-S-access3	Fa0/4	40	M-HR PC	0001.C91C.D35B
M-S-access4	Fa0/3	30	M-finance PC	0010.1135.6484
M-S-access4	Fa0/4	50	M-sales PC	00E0.B0A6.5A1D
M-S-DMZ	Fa0/1, Fa0/2	15	DNS server, Email/Web server	No port security (server ports, PortFast)
D-switch1	Fa0/1 / Fa0/2	110 / 120	D-operations / D-IT & Admin	0001.9766.B424 / 0000.0CA9.4860
D-switch2	Fa0/1 / Fa0/2	130 / 140	D-guest / D-Finance & Sales	00D0.583E.CB78 / 0002.1656.B04A
K-switch1	Fa0/1 / Fa0/2	210 / 220	K-operations / K-IT & Admin	0001.6358.ED8A / none recorded
K-switch2	Fa0/1 / Fa0/2	230 / 240	K-guest / K-Finance & Sales	0002.17E4.DE7D / 0090.0C4E.604B

All remaining ports on every switch are assigned to the parking VLAN (58 at HQ, 99 at the branches) and administratively shut down, so an unauthorised device plugged into a wall socket receives no network service.

3.4 Native VLAN and VTP mode

To manage the switching infrastructure across the local site efficiently, a consolidated VLAN Trunking Protocol (VTP) domain was established alongside standard IEEE 802.1Q trunking.

VTP Domain & Mode: The primary distribution switch (M-S-core1) operates as the VTP Server, while the secondary access switches operate as VTP Clients. This architecture allows for centralized VLAN administration and ensures synchronized VLAN databases across all trunk links without manual database entries on each switch.

Trunking & Native VLAN: All inter-switch links are configured as 802.1Q trunks, carrying tagged traffic for all operational VLANs (such as Management VLAN 70). The default untagged traffic is consistently managed across all trunk ends on Native VLAN 1, maintaining operational stability and full end-to-end connectivity without trunk misconfigurations.

```
M-S-core1#show vtp status
VTP Version capable             : 1 to 2
VTP version running             : 2
VTP Domain Name                 : AN-tech.com
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 000D.BDBD.E820
Configuration last modified by 0.0.0.0 at 3-6-93 12:13:20
Local updater ID is 192.168.11.102 on interface Vl70 (lowest numbered VLAN
interface found)

Feature VLAN :
--------------
VTP Operating Mode                : Server
Maximum VLANs supported locally   : 255
Number of existing VLANs          : 14
Configuration Revision            : 97
MD5 digest                        : 0x33 0x1A 0x73 0x22 0x4A 0x4E 0x44 0x64
                                    0xDE 0x1A 0xC4 0xD4 0x3D 0x37 0x64 0x94
M-S-core1#show interfaces trunk
```

Figure 5: *Evidence – `show vtp status`*

```
M-S-core1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      1
Fa0/1       auto         n-802.1q       trunking      1
Fa0/2       auto         n-802.1q       trunking      1
Fa0/6       auto         n-802.1q       trunking      1
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Po1         10,20,30,40,50,60,70,80
Fa0/1       1-1005
Fa0/2       1-1005
Fa0/6       1-1005
Gig0/1      10,20,30,40,50,60,70,80

Port        Vlans allowed and active in management domain
Po1         10,20,30,40,50,60,70,80
Fa0/1       1,10,20,30,40,50,58,60,70,80
Fa0/2       1,10,20,30,40,50,58,60,70,80
Fa0/6       1,10,20,30,40,50,58,60,70,80
Gig0/1      10,20,30,40,50,60,70,80

Port        Vlans in spanning tree forwarding state and not pruned
Po1         10,20,30,40,50,60,70,80
Fa0/1       1,10,20,30,40,50,58,60,70,80
Fa0/2       1,10,20,30,40,50,58,60,70,80
Fa0/6       1,10,20,30,40,50,58,60,70,80
Gig0/1      10,20,30,40,50,60,70,80

M-S-core1#
```

Figure 6: *Evidence – `show interfaces trunk`*

3.5 Spanning Tree configuration

HQ switches run **Rapid PVST+** for fast reconvergence on the dual-uplink access layer; the Dubai, Kuwait and DMZ switches run **PVST+**, which is adequate for their small two-switch topologies. The two modes interoperate on the DMZ trunk. In every switch pair, the root bridge is chosen deliberately by lowering the priority, instead of letting the lowest MAC address win.

Table 22: *STP configuration per site*

Site	Mode	Root bridge (priority)	Secondary (priority)	Additional protection
Manama	Rapid PVST+	M-S-core1: 24576 for VLANs 10–80	M-S-core2	`portfast default` on access switches; `bpduguard default` on M-S-access3 and M-S-access4
Dubai	PVST+	D-switch1: 24576 (VLANs 110–140)	D-switch2: 28672	PortFast on host ports
Kuwait	PVST+	K-switch2: 20480 (VLANs 210–240)	K-switch1: 24576	PortFast on host ports

```
M-S-core1#show spanning-tree vlan 10
VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    24586
             Address     0001.C94D.305D
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24586  (priority 24576 sys-id-ext 10)
             Address     0001.C94D.305D
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/2            Desg FWD 19        128.2    P2p
Fa0/1            Desg FWD 19        128.1    P2p
Fa0/6            Desg FWD 19        128.6    P2p
Gi0/1            Desg FWD 4         128.25   P2p
Po1              Desg FWD 12        128.27   P2p

M-S-core1#
```

Figure 7: *Evidence – `show spanning-tree vlan 10` on M-S-core1 (root bridge) and an access switch (root port + alternate port)*

```
M-S-access4#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gig0/1, Gig0/2
10   Administrators                   active
20   Management                       active
30   Finance                          active    Fa0/3
40   HR                               active
50   Sales                            active    Fa0/4
58   unused-ports                     active    Fa0/1, Fa0/5, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
60   Compliance                       active
70   Operational-IT                   active
80   Wireless&Guests                  active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
M-S-access4#
```

Figure 8: *Evidence – `show vlan brief` on M-S-access4*

4. Network Security (Requirement 3)

4.1 Theory: threats and the controls used against them

Security in this design follows **defence in depth**: several independent controls protect each asset, so that the failure or bypass of one control does not expose the network. Controls are applied at the edge, in the distribution layer, on the switch ports, on the WAN links and on the management plane. Each control is explained in its own section; this section shows how they fit together.

Table 24: *Threat and mitigation matrix*

Threat	Where it appears	Control	Detailed in
Reconnaissance (ping sweeps) of public servers	DMZ web, email and DNS servers	ICMP denied to the DMZ servers (`EDGE_SECURITY_ACL`)	Section 9
Guests reaching internal data	Wireless & guest VLANs at all sites	Guest VLANs isolated from `192.168.0.0/16`; dedicated VLAN; separate SSID	Sections 6 and 9
Unauthorised administration of devices	VTY lines on every router/switch	SSH v2 only, local users, `access-class` ACL, banner, exec-timeout	Section 10
Rogue devices on access ports (MAC spoofing, CAM flooding)	Access switches	Port security with sticky MAC, unused ports shut down	Section 11
Rogue switches / STP manipulation	Access switches	PortFast + BPDU Guard, deterministic root bridge	Section 3
Eavesdropping on Wi-Fi	Wireless VLAN	WPA2/AES encryption	Section 6
Internal address exposure	Internet edge	NAT/PAT hides all private addressing	Section 8
Spoofed WAN neighbours	Serial WAN links	PPP with CHAP authentication	Section 7
Gateway failure / DoS by outage	Default gateway, links	HSRP, EtherChannel, redundant WAN	Sections 5, 12, 13
Credential theft from configuration files	Running/startup config	`enable secret`, `username ... secret`, `service password-encryption`	Section 10

4.2 Security controls implemented per device

The matrix summarises which controls are present in the saved configurations. A tick means the control is present; a dash means it is absent or incomplete (details are discussed in the relevant section and in Section 17).

Table 25: *Security control matrix*

Device	SSH v2	VTY ACL (access-class 10)	Secret passwords	Password encryption	Banner	Port security	ACL / NAT
M-R-edge	Yes	–	Yes	Yes	Yes	n/a	NAT_ACL, EDGE_SECURITY_ACL (not yet applied)
M-R-dist1 / dist2	Yes	Yes	Yes	Yes	Yes	n/a	Standard ACL 10
M-S-core1	Yes	Yes	Yes	Yes	Yes	n/a	ACL 10
M-S-access1	Yes	Yes	Yes	Yes	Yes	Partial	ACL 10, exec-timeout 5
M-S-access2	Yes	Yes	Yes	– (disabled)	Yes	Yes	ACL 10
M-S-access3 / 4	Yes	Yes	Yes	Yes	Yes	Yes + BPDU Guard	ACL 10
M-S-DMZ	Yes	Yes	Yes	– (disabled)	Yes	n/a (server ports)	ACL 10
ISP	–	–	Yes	Yes	–	n/a	– (simulated provider)
D-R-edge / K-R-edge	Yes	Config defined (ACL 10)	Yes	Yes	Yes	n/a	ACL 10
D-R-dist / K-R-dist	Yes	Yes	Yes	Yes	Yes	n/a	ACL 10
Branch switches	Yes	Yes	Yes	Yes	Yes	Yes (host ports)	ACL 10

5. Network Redundancy and High Availability (Requirement 4)

5.1 Theory: single points of failure and redundancy mechanisms

A **single point of failure (SPOF)** is any component whose failure stops a service for users. High availability is achieved by identifying each SPOF and providing an alternative at every layer: gateway, switch, link and WAN. The mechanisms used in this design are summarised below, and each is described in detail in its own section.

Table 26: *Redundancy mechanisms used*

Layer	Problem	Mechanism	Section
Default gateway	Hosts have one gateway address; if the router fails they lose all off-subnet access	First-hop redundancy: **HSRP** with a virtual IP and MAC shared by two routers	12
Layer 2 links	One cable failure isolates a switch; two cables create loops	**EtherChannel** (bundle of links) + **STP/RSTP** (blocks loops, restores on failure)	13, 3
Access to core	One uplink per access switch is a SPOF	Each access switch has two uplinks, one to each core switch; STP blocks one	3
WAN	A single link between sites	Triangle topology with **OSPF** automatic rerouting	2, 7
Routing	Manual routes do not adapt	Dynamic routing (OSPF) with fast convergence	2

5.2 SPOF analysis of the implemented design

Table 27: *SPOF analysis*

Component	Redundant?	Mechanism / comment
HQ default gateway (VLANs 10–80)	Yes	M-R-dist1 and M-R-dist2 share one virtual IP per VLAN (HSRP)
HQ core switches	Yes	Two core switches joined by a two-link EtherChannel; every access switch is dual-homed
Core-to-dist links	Yes	Each core switch is attached to a different distribution router
HQ Layer 2 trunk/EtherChannel	Yes	Two member links; the port-channel keeps working if one fails
Inter-site WAN	Yes	Three serial links in a triangle; OSPF reroutes in seconds
Branch LAN switches	Partly	Two switches joined by EtherChannel; the router connection uses a single trunk
Branch inter-VLAN router (D-R-dist, K-R-dist)	No	Single router per branch (acceptable for a small branch; second router + HSRP is the upgrade path)
Branch WAN edge router	No	One edge router per branch, but its two serial links provide path redundancy
M-R-edge (Internet and WAN hub)	No	Only Internet gateway and the transit point between HQ and branches. Upgrade path: second edge router + second ISP link
ISP link	No	Single link to a single provider
DNS / Email / Web servers	No	One of each in the DMZ; cluster or secondary server is the upgrade path
DHCP service	No	Hosted only on M-R-dist1 at HQ (see Section 14)

The analysis is deliberately honest: the redundancy requirements are met for the headquarters LAN and for the WAN, while the remaining single points are documented together with an upgrade path. This is a normal cost-versus-risk decision for an organization of this size.

5.3 Failover test plan

Table 28: *Redundancy tests*

#	Failure introduced	Expected behaviour	Evidence
4.1	Shut `Gi0/2` of M-R-dist1 (or power it off) while pinging the VLAN gateway	Standby router becomes active; pings continue after 1–3 lost replies	`show standby brief` on M-R-dist2
4.2	Disconnect one EtherChannel member	Port-channel stays up with one link; no outage	`show etherchannel summary`
4.3	Disconnect an access-switch uplink	RSTP unblocks the alternate uplink	`show spanning-tree`
4.4	Disconnect `10.1.0.0/30` (HQ–Dubai) and ping Dubai from HQ	Traffic uses HQ→Kuwait→Dubai; tracert shows an extra hop	`tracert`, `show ip route`

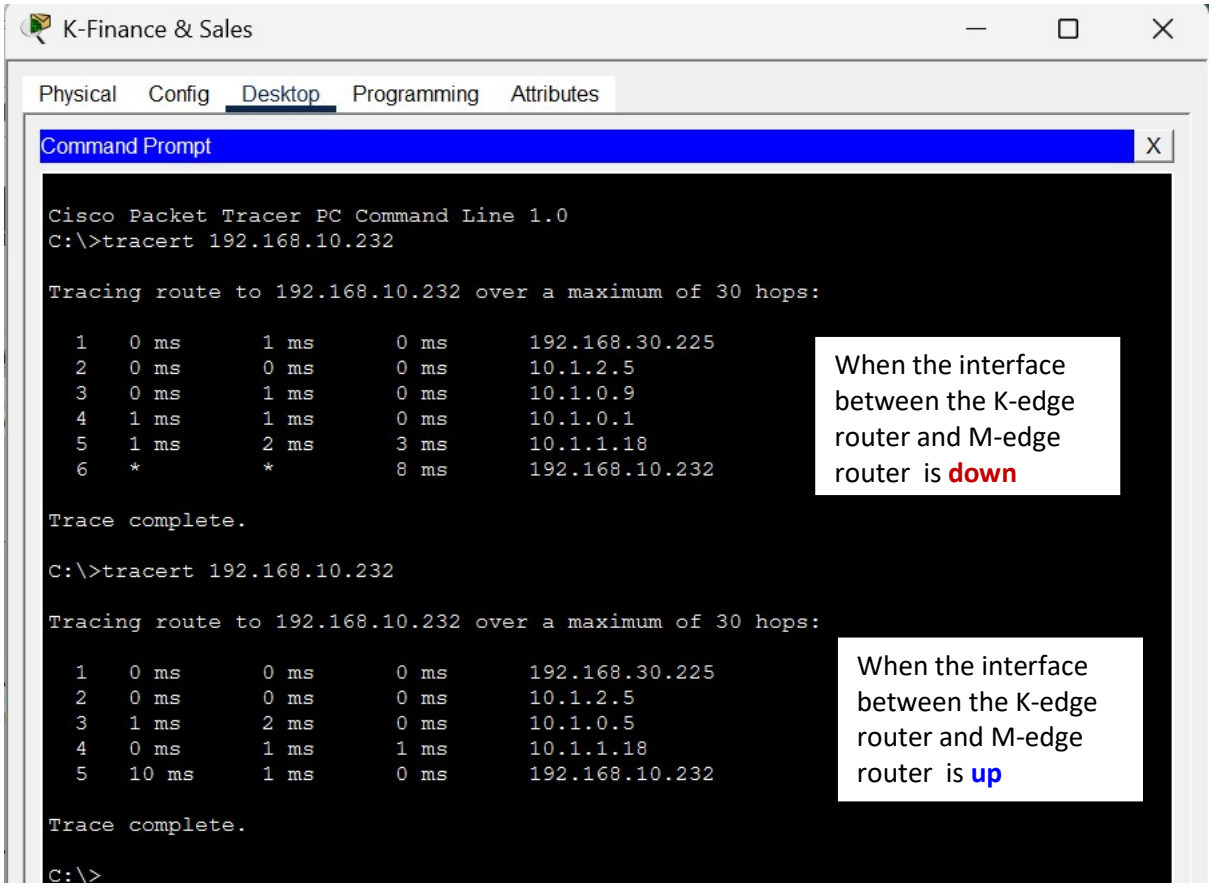


Figure 9: *Evidence – Ping with a failed link and the resulting `tracert` path via Kuwait*

6. Secure Wireless Network (Requirement 5)

6.1 Theory: wireless security standards

Radio signals can be received by anyone within range, so a wireless network must authenticate users and encrypt data itself; it cannot rely on physical access control. The standards evolved because each earlier one was broken.

Table 29: *Comparison of wireless security protocols*

Protocol	Year	Encryption	Authentication	Status
WEP	1997	RC4 with 40/104-bit static key, 24-bit IV	Shared key	**Broken** – key recovered in minutes; must not be used
WPA	2003	TKIP (RC4 with per-packet key mixing)	PSK or 802.1X	Deprecated interim fix for old hardware; vulnerable to several attacks
WPA2	2004	AES-CCMP (128-bit)	PSK (Personal) or 802.1X/RADIUS (Enterprise)	**Current baseline**; secure with a strong passphrase
WPA3	2018	AES-CCMP / GCMP-256 (Enterprise 192-bit)	SAE (Personal) or 802.1X; protected management frames mandatory	Strongest; resists offline dictionary attacks and gives forward secrecy; requires newer devices

Table 30: *TKIP versus AES*

Feature	TKIP	AES (CCMP)
Cipher	RC4 stream cipher (legacy)	AES block cipher in counter mode
Integrity	Michael MIC (weak)	CBC-MAC (strong)
Used by	WPA	WPA2 and WPA3
Known attacks	Beck–Tews, Michael countermeasure abuse	None practical against a strong key
Performance	Limits 802.11n/ac to low speeds	Full speed; hardware accelerated
Decision	Rejected	**Selected**

6.2 Selection and justification

WPA2-Personal with AES-CCMP is the security solution selected for the HQ wireless network. WEP and WPA/TKIP are rejected because they can be broken. WPA3 is the best technology available, but it is not offered by the simulated wireless router and many existing guest devices do not support it yet; it is therefore the recommended migration target. WPA2-Enterprise (802.1X with a RADIUS server) would give per-user credentials, but it requires a RADIUS server and certificate infrastructure that is disproportionate for a guest and visitor network. The risk of a shared passphrase is reduced by the additional controls below.

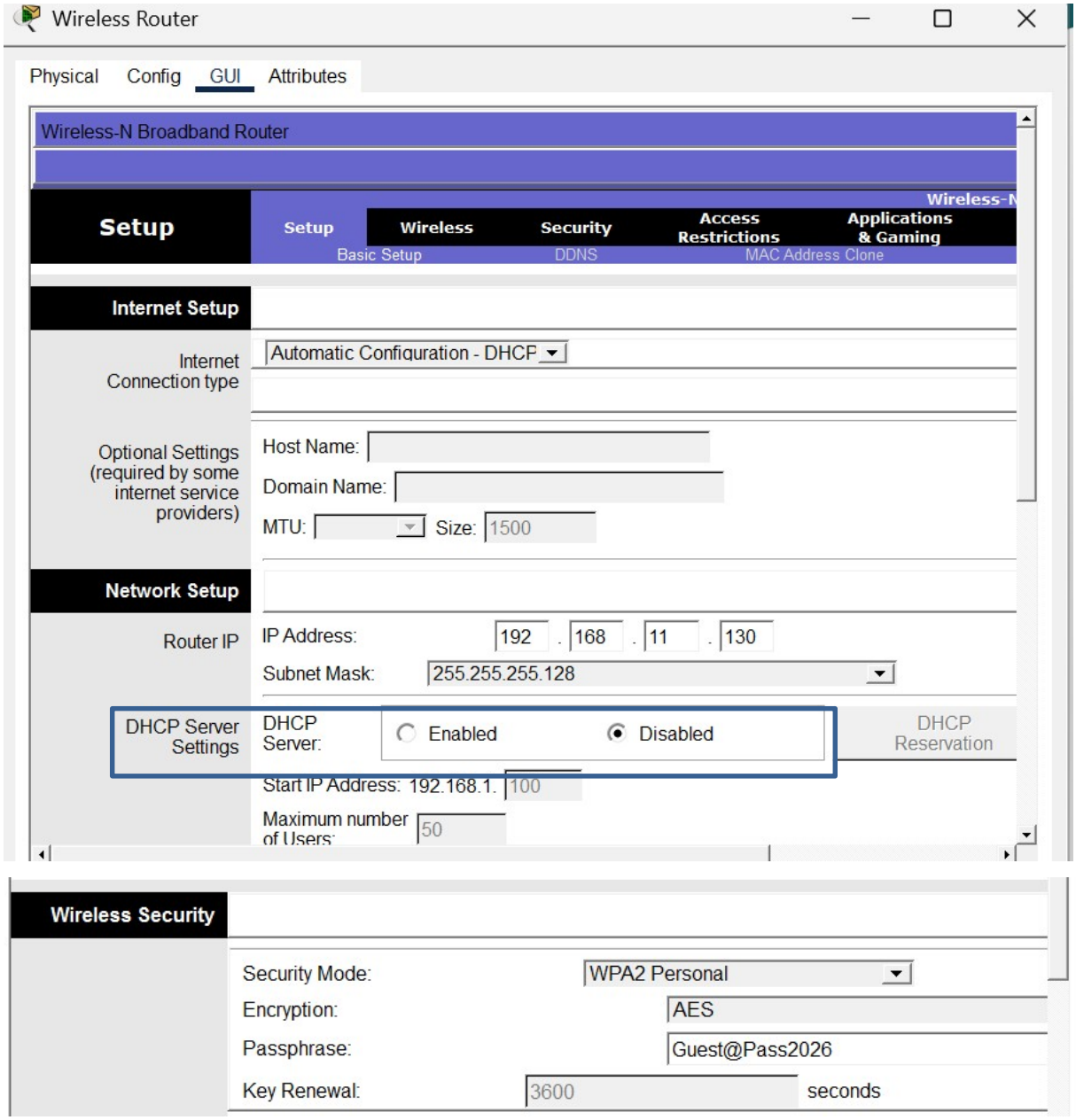
6.3 Wireless network design at Manama HQ

Table 31: *Wireless design parameters*

Parameter	Value	Source / justification
SSID	`AN-guests`	Guest/wireless network name used in the design
VLAN and subnet	VLAN 80, `192.168.11.128/25` (126 hosts)	Dedicated segment, separate from every wired department
Gateway	`192.168.11.129` (HSRP virtual IP of M-R-dist1/2)	Redundant gateway like any other VLAN

Parameter	Value	Source / justification
DHCP	Pool `VLAN80_POOL` on M-R-dist1, DNS `10.1.1.2`	.129–.140 excluded for gateway, router and static devices
Switch attachment	M-S-access1 `Fa0/4` (access VLAN 80)	Wireless router attached to a dedicated access port
Security mode	WPA2-Personal, AES	Selected above
Passphrase	Guest@Pass2026	Mitigates brute-force and sharing

The dedicated VLAN means wireless users share a broadcast domain only with each other. All their traffic must cross a Layer 3 boundary (the distribution routers) before it can reach any wired department, which is the point where policy is enforced.



The wireless router connected via its LAN port with DHCP disabled. This ensures that wireless clients do not receive isolated local addresses (like 192.168.0.x) via a secondary NAT. Instead, they bridge directly into VLAN 80, acquiring their IP addresses dynamically from the central VLAN80_POOL (192.168.11.128/25), maintaining consistent network segmentation and security policies.

6.4 Wireless tests

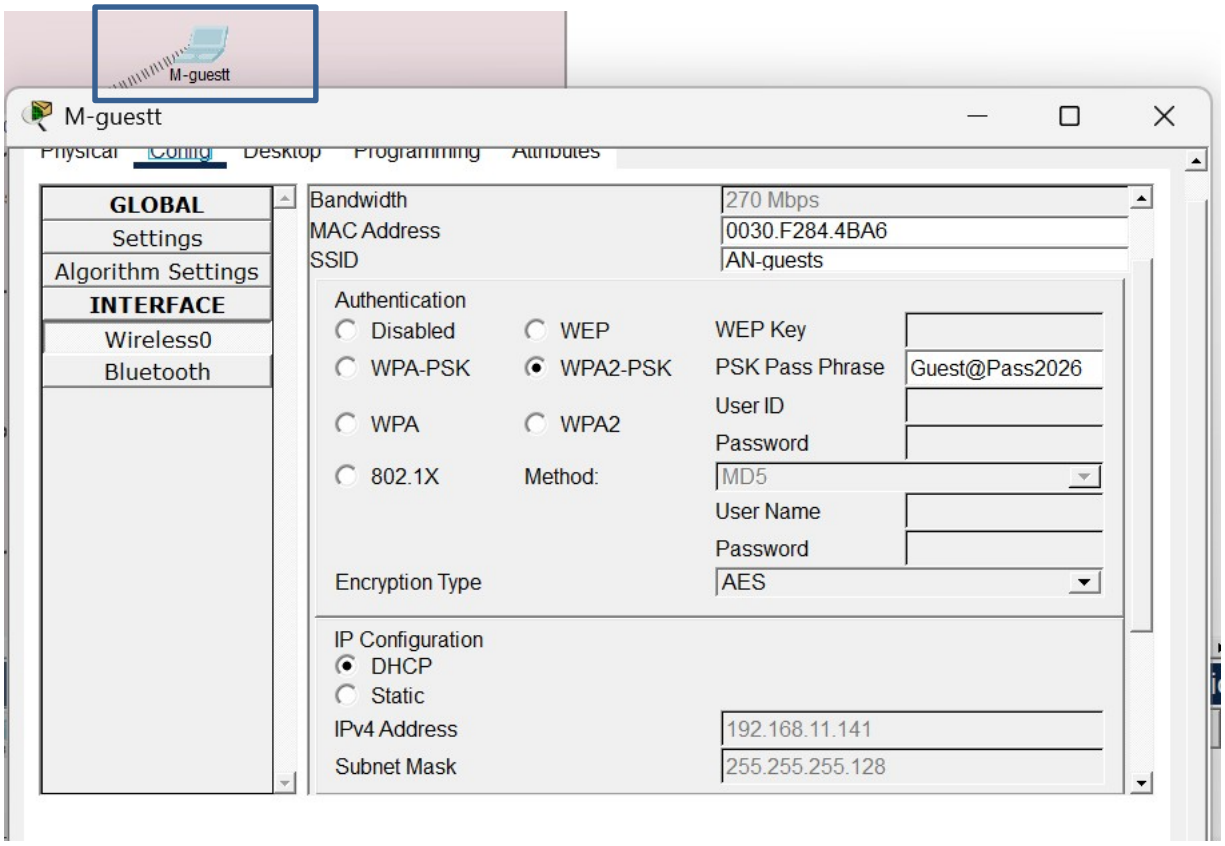


Figure 10: *Evidence – Laptop wireless association and IP configuration (`ipconfig`) in VLAN 80*

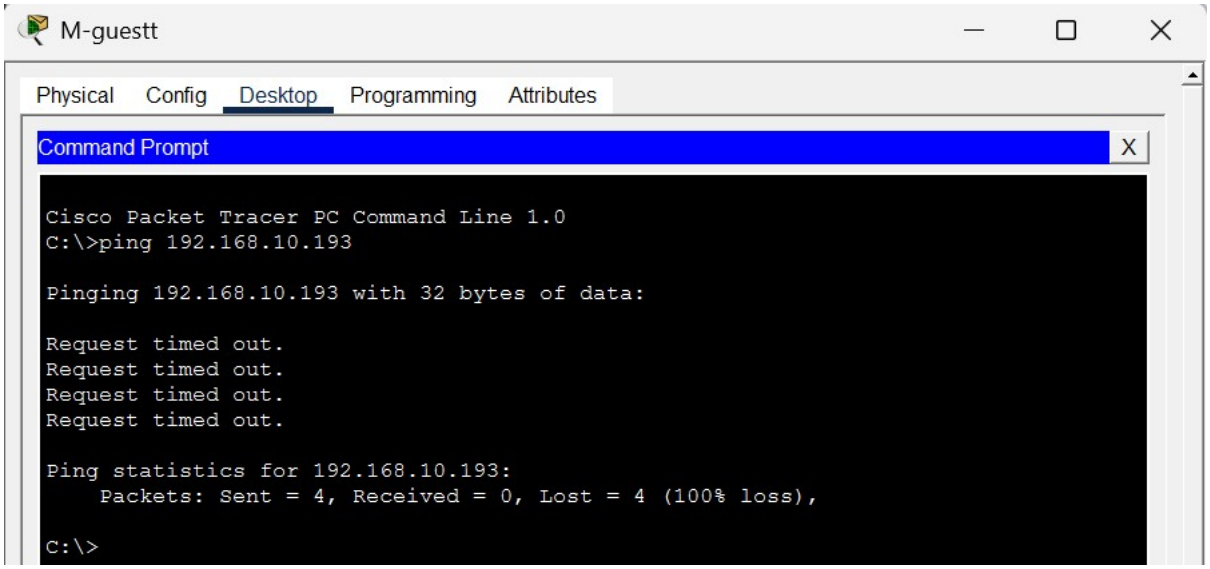


Figure 11: *Evidence – Failed ping from the wireless laptop to a Finance host*

7. WAN Design (Requirement 6)

7.1 Theory: WAN technology (PPP)

A **Wide Area Network** connects sites that are too far apart for a LAN, normally using circuits leased from a service provider. A **point-to-point link** (leased line) gives a dedicated, private path between exactly two sites.

Point-to-Point Protocol (PPP)

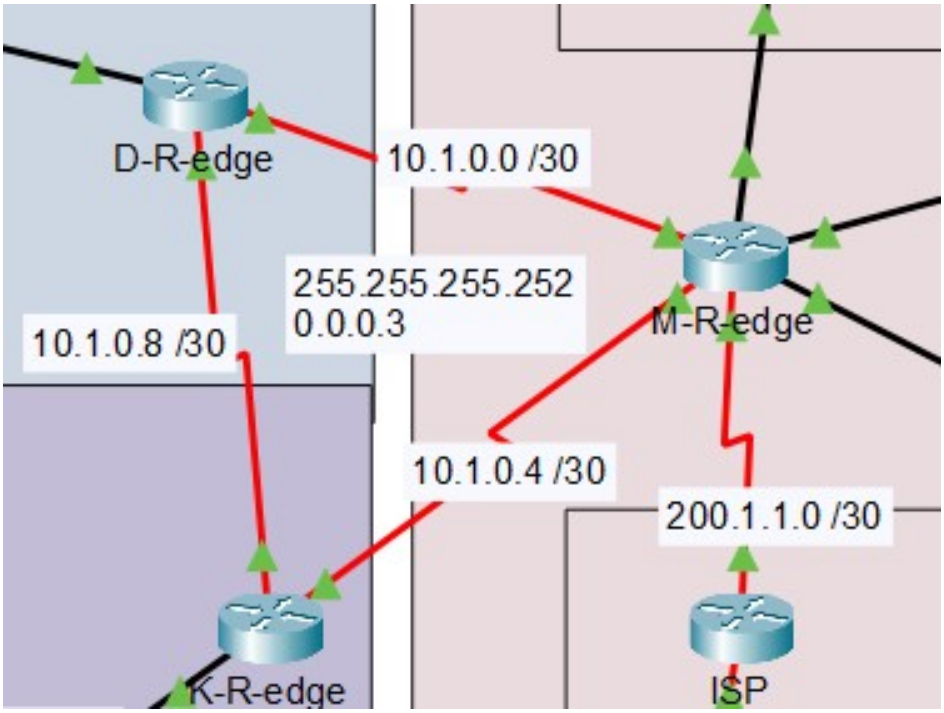
PPP (RFC 1661) is an open data-link protocol for point-to-point links. Unlike Cisco's default HDLC encapsulation it is standards-based and provides several features that are valuable in a business network.

Table 35: *PPP features*

Feature	Function	Benefit
LCP (Link Control Protocol)	Establishes, configures and tests the link; negotiates options	Automatic link set-up and loop detection
NCP (Network Control Protocols)	One per network protocol (IPCP for IPv4, IPv6CP)	Multiprotocol support
Authentication	PAP or CHAP before the link comes up	Only trusted routers can connect
Link quality monitoring	Measures errors and can close a poor link	Early detection of bad circuits
Multilink PPP	Bundles several serial links	More bandwidth (future)
Compression / error detection	Optional	Efficiency and integrity

7.2 Selected WAN design

The three sites are connected by **three serial point-to-point links forming a triangle**: HQ–Dubai, HQ–Kuwait and Dubai–Kuwait. A full mesh of three sites needs only three links, and it provides two paths between every pair of sites, which satisfies the requirement for an alternative WAN path without any additional equipment. Every link is a /30 from `10.1.0.0/16`.



7.3 PPP and CHAP authentication

To enforce authentication and protect inter-site WAN communications across the network, Point-to-Point Protocol (PPP) with Challenge Handshake Authentication Protocol (CHAP) was deployed across all WAN serial interfaces linking the Manama headquarters (M-R-edge), Riyadh branch (D-R-edge), and Kuwait branch (K-R-edge).
Authentication Protocol: ppp authentication chap configured globally on all active inter-router serial interfaces.
Credential Mapping: Each router authenticates its remote peer using the peer's exact hostname as the username (e.g., M-R-edge, D-R-edge, and K-R-edge).
Shared Secret: A standardized, strong shared password (**AntechRouters**) was configured across all three routers to ensure smooth mutual authentication and dynamic session negotiation.

```
M-R-edge#show interfaces serial 0/1/0
Serial0/1/0 is up, line protocol is up (connected)
  Hardware is HD64570
  Internet address is 10.1.0.1/30
  MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation PPP, loopback not set, keepalive set (10 sec)
  LCP Open
  Open: IPCP, CDPCP
  Last input never, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: weighted fair
  Output queue: 0/1000/64/0 (size/max total/threshold/drops)
     Conversations  0/0/256 (active/max active/max total)
     Reserved Conversations 0/0 (allocated/max allocated)
     Available Bandwidth 1158 kilobits/sec
  5 minute input rate 22 bits/sec, 0 packets/sec
  5 minute output rate 37 bits/sec, 0 packets/sec
     756 packets input, 53376 bytes, 0 no buffer
     Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     724 packets output, 51960 bytes, 0 underruns
     0 output errors, 0 collisions, 0 interface resets
     0 output buffer failures, 0 output buffers swapped out
     0 carrier transitions
     DCD=up  DSR=up  DTR=up  RTS=up  CTS=up

M-R-edge#
06:15:14: %OSPF-5-ADJCHG: Process 1, Nbr 10.1.2.5 on Serial0/2/0 from LOADING to
FULL, Loading Done
```

Figure 12: *Evidence – `show interfaces serial 0/1/0` with PPP/CHAP*

8. NAT and PAT (Requirement 7)

8.1 Theory

Network Address Translation (NAT, RFC 3022) rewrites IP addresses as packets cross a router so that private hosts can use public addresses. It slows the exhaustion of IPv4 and hides the internal address plan. A router interface is marked `ip nat inside` (towards the private network) or `ip nat outside` (towards the Internet).

Table 39: *Types of NAT*

Type	Mapping	Use	Limitation
Static NAT	One private address ↔ one public address, permanent	Servers that must be reachable from the Internet	Consumes one public address per server
Dynamic NAT	Private addresses ↔ a pool of public addresses, first come first served	Few hosts, several public addresses	Pool exhaustion drops new sessions
PAT (overload)	Many private addresses ↔ one public address, distinguished by port	Whole organisation with one public address	Port exhaustion at extreme scale; breaks some protocols

Table 40: *NAT address terminology with examples from this network*

Term	Meaning	Example in this design
Inside local	Private address of an inside host as seen inside	`192.168.10.20` (a Finance PC)
Inside global	Public address that represents the inside host to the outside	`200.1.1.1` (address of `M-R-edge Se0/1/1`)
Outside local	Address of an outside host as seen from inside	`200.1.1.2` (ISP)
Outside global	Real address of the outside host	`200.1.1.2` (ISP)

Advantages: conserves public addresses, hides the internal topology, lets the organisation change its provider without renumbering. **Disadvantages:** breaks end-to-end addressing and traceability, some protocols embed IP addresses and fail through NAT, adds processing delay, complicates IPsec and logging.

8.2 Design decision and implementation

The ISP link provides **a single public address** to the organisation (`200.1.1.1/30` on `M-R-edge`). Static or dynamic NAT with a pool therefore cannot serve many users, so **PAT with interface overload** was selected: every private source address is translated to `200.1.1.1` with a unique source port. NAT is performed only on `M-R-edge`, the single point where the Internet is reached; the branches and the distribution routers do not run NAT, which keeps the translation table in one place for monitoring.

```
ip nat inside source list NAT_ACL interface Serial0/1/1 overload

ip access-list standard NAT_ACL
 permit 192.168.0.0 0.0.255.255
 permit 10.1.0.0 0.0.255.255

interface Serial0/1/1
 ip address 200.1.1.1 255.255.255.252
 ip nat outside
```

Table 41: *NAT interface roles on M-R-edge*

Interface	Address	NAT role	Network
Gi0/0	10.1.1.17/30	inside	Link to M-R-dist1 (all HQ VLANs)
Gi0/2	10.1.1.21/30	inside	Link to M-R-dist2 (all HQ VLANs)
Gi0/1.15	10.1.1.1/28	inside	DMZ (DNS, Email/Web)
Gi0/1.70	192.168.11.97/27	inside	Administrators VLAN (see address note, Section 2.3)
Se0/1/0	10.1.0.1/30	inside	WAN to Dubai (and everything behind it)
Se0/2/0	10.1.0.5/30	inside	WAN to Kuwait (and everything behind it)
Se0/1/1	200.1.1.1/30	**outside**	ISP / Internet

The ACL `NAT_ACL` selects which sources may be translated: all user networks (`192.168.0.0/16`, covering HQ, Dubai and Kuwait) and the infrastructure range (`10.1.0.0/16`, covering DMZ and router-to-router addresses). A default route `ip route 0.0.0.0 0.0.0.0 200.1.1.2` sends all non-local traffic to the ISP, and `default-information originate` shares it with the rest of the network through OSPF.

8.4 Static NAT (Port Forwarding) for Public DMZ Services

To satisfy Requirement 7 and expose internal DMZ services to external networks using a single public IP address (200.1.1.1), Port-based Static NAT (Port Forwarding) was configured on the main edge router (M-R-edge). Inbound traffic targeting 200.1.1.1 is mapped directly to internal DMZ servers according to specific service port numbers:

DNS Server (10.1.1.2): Ports UDP 53 and TCP 53 forwarded for domain resolution services.

Web & Mail Server (10.1.1.3): Port TCP 80 (HTTP) forwarded for public Web access.

Port TCP 25 (SMTP) and Port TCP 110 (POP3) forwarded for incoming and outgoing email traffic.

8.4 NAT verification

Table 42: *NAT tests*

#	Test	Expected result
7.1	Ping `200.1.1.2` from an HQ PC, then `show ip nat translations` on M-R-edge	ICMP entry: inside local = PC address, inside global = `200.1.1.1`
7.2	Repeat from a Dubai or Kuwait PC	Same translation; path via HQ
7.3	`show ip nat statistics`	Inside/outside interfaces listed, hits increasing

```
M-R-edge#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
udp 200.1.1.1:53       10.1.1.2:53        ---                ---
tcp 200.1.1.1:110      10.1.1.3:110       ---                ---
tcp 200.1.1.1:25       10.1.1.3:25        ---                ---
tcp 200.1.1.1:53       10.1.1.2:53        ---                ---
tcp 200.1.1.1:80       10.1.1.3:80        ---                ---

M-R-edge#show ip nat statistics
Total translations: 5 (5 static, 0 dynamic, 5 extended)
Outside Interfaces: Serial0/1/1
Inside Interfaces: GigabitEthernet0/0 , GigabitEthernet0/2 , Serial0/1/0 ,
Serial0/2/0 , GigabitEthernet0/1.15 , GigabitEthernet0/1.70
Hits: 12  Misses: 12
Expired translations: 12
Dynamic mappings:
M-R-edge#
```

Figure 13: *Evidence – `show ip nat translations` and `show ip nat statistics` on M-R-edge*

9. Access Control List Design (Requirement 8)

9.1 Theory: types, direction, placement and least privilege

An **Access Control List (ACL)** is an ordered list of permit/deny statements that a router evaluates top-down; the first match decides, and an invisible `deny ip any any` ends every ACL. ACLs filter traffic on interfaces and restrict management access on VTY lines. The **principle of least privilege** requires that every user and device receives only the access that its role needs.

Table 43: *ACL types*

Type	Matches	Number range	Placement rule
Standard	Source IP only	1–99, 1300–1999	As close to the **destination** as possible (cannot see the destination)
Extended	Source, destination, protocol, ports	100–199, 2000–2699	As close to the **source** as possible, to drop unwanted traffic early
Numbered	Identified by number	As above	Older style; editing requires sequence numbers
Named	Identified by a name	Any	Preferred: readable, individual lines can be edited and deleted

- **Inbound** ACLs are checked before the routing decision; denied packets cost almost no router resources. **Outbound** ACLs are checked after routing and do not filter packets generated by the router itself.
- An ACL must end with an explicit permit for the traffic that should still work, because of the implicit deny.
- Wildcard masks are the inverse of subnet masks: `/27` → `0.0.0.31`, `/26` → `0.0.0.63`, `/25` → `0.0.0.127`, `/16` → `0.0.255.255`.

9.2 Policy summary

Table 44: *Access policy implemented*

Policy goal	ACL / mechanism	Device
Only administrators may manage devices remotely	Standard ACL 10 on VTY lines (`access-class 10 in`)	All routers/switches (Section 10)
Internet access for all internal users	Named standard ACL `NAT_ACL` selecting traffic for PAT	M-R-edge
Hide DMZ servers from ping scans	`EDGE_SECURITY_ACL` rules 1–2	M-R-edge (Gi0/1.15 outbound
Guests cannot reach internal networks but can use the Internet	`EDGE_SECURITY_ACL` rules 3–5 and guest permits	Distribution Routers (M-R-dist1/2, D-R-dist, K-R-dist)
Public access only to approved services	DMZ is the only segment addressed by the published services; internal networks are never published	M-R-edge NAT
No direct access to sensitive internal networks	Sensitive VLANs have no NAT/static mapping; guest VLANs denied	M-R-edge, dist1,2

9.3 EDGE_SECURITY_ACL

The extended ACL below is defined on `M-R-edge`. Its rules are ordered so that specific denies come before the general permits.

```
ip access-list extended EDGE_SECURITY_ACL
 deny icmp any host 10.1.1.2
 deny icmp any host 10.1.1.3
 deny ip 192.168.11.128 0.0.0.127 192.168.0.0 0.0.255.255
 deny ip 192.168.20.192 0.0.0.31 192.168.0.0 0.0.255.255
 deny ip 192.168.30.192 0.0.0.31 192.168.0.0 0.0.255.255
 permit ip 192.168.0.0 0.0.255.255 any
 permit ip 192.168.11.128 0.0.0.127 any
 permit ip 192.168.20.192 0.0.0.31 any
 permit ip 192.168.30.192 0.0.0.31 any
```

Table 45: *Rule-by-rule explanation*

Rule	Statement	Purpose and justification
1	deny icmp any host 10.1.1.2	Blocks ping to the DNS server. ICMP echo is the first tool of reconnaissance; DNS itself (UDP/TCP 53) still works.
2	deny icmp any host 10.1.1.3	Same protection for the Email/Web server (HTTP/HTTPS/SMTP unaffected).
3	deny ip 192.168.11.128/25 → 192.168.0.0/16	Manama guest/wireless VLAN 80 may not reach any internal network at any site.
4	deny ip 192.168.20.192/27 → 192.168.0.0/16	Dubai guest VLAN 130 may not reach internal networks.
5	deny ip 192.168.30.192/27 → 192.168.0.0/16	Kuwait guest VLAN 230 may not reach internal networks.
6	permit ip 192.168.0.0/16 any	All other user networks may reach the DMZ, other sites and the Internet.
7–9	permit ip <guest subnet> any	Documents the intent that guests may reach the Internet and DMZ. Because guest sources also fall inside `192.168.0.0/16`, rule 6 already permits them after rules 3–5 have removed internal destinations, so rules 7–9 are never reached; they are harmless and can be removed for clarity.
implicit	deny ip any any	Anything else (including sources from `10.1.0.0/16`) is dropped, so the ACL must **not** be placed where routing-protocol traffic (OSPF, HSRP) arrives.

9.4 Standard ACL 10 for management access

Table 46: *ACL 10 (VTY access-class) by site*

Site	Permitted source	Meaning
Manama (dist1/2, core, access, DMZ switch)	`permit 192.168.11.96 0.0.0.31`	VLAN 70 Administrators only
Dubai (D-R-dist, D-R-edge, D-switch1/2)	`permit 192.168.20.128 0.0.0.63`	VLAN 120 IT & Admin only
Kuwait (K-R-dist, K-R-edge, K-switch1/2)	`permit 192.168.30.128 0.0.0.63`	VLAN 220 IT & Admin only

This is a standard ACL because only the source of the SSH session needs to be matched, and it is used as an `access-class` so it filters only the router's own VTY lines and not transit traffic. A consequence of the design is that administration is **local to each site**: HQ administrators cannot SSH to branch devices unless the HQ Administrators subnet is added to the branch ACLs.

9.5 GUEST_IN ACL

The extended access control list below is applied inbound on the guest sub-interfaces at the distribution layer (M-R-dist1/2, D-R-dist, and K-R-dist). Its rules prioritize network bootstrapping and DMZ/Internet clearance while enforcing strict isolation from internal corporate networks.

```
ip access-list extended GUEST_IN
 permit udp any any eq bootps
 deny ip <guest_subnet> <wildcard> 192.168.0.0 0.0.255.255
 deny ip <guest_subnet> <wildcard> 10.1.0.0 0.0.255.255
 permit ip any any
```

Table 45: *Rule-by-rule explanation*

Rule	Statement	Purpose and justification
1	permit udp any any eq bootps	Permits DHCP IP Allocation: Essential for new guest devices to acquire network parameters (IP, Subnet Mask, Default Gateway, DNS) via BOOTP/DHCP protocol before authentication, as the client source IP is initially 0.0.0.0.
2	deny ip <guest_subnet> 192.168.0.0/16	Blocks Access to Internal Corporate Networks: Drops all IP packets originating from the guest subnet targeting private corporate user/management subnets (192.168.x.x), enforcing strict intra-site and inter-site guest isolation.
3	deny ip <guest_subnet> 10.1.0.0/16	Blocks Access to Internal Infrastructure & Services: Drops guest traffic targeting internal router subnets, core loopbacks, and non-public DMZ segments residing within the 10.1.x.x address space
4	permit ip any any	Permits Internet Outbound Access: Clears all remaining guest traffic to traverse default dynamic NAT/PAT translation at the edge router (M-R-edge) for unrestricted public web and Internet browsing
5	deny ip 192.168.30.192/27 → 192.168.0.0/16	Kuwait guest VLAN 230 may not reach internal networks.
implicit	deny ip any any	Default Drop Action: Catches and drops any unauthorized packets that do not match the above explicit conditions.

9.7 NAT_ACL

To enable outbound Internet connectivity for all internal corporate subnets across the HQ and regional branches while maintaining explicit control over translation boundaries, a Standard Access Control List (NAT_ACL) was deployed on M-R-edge.

Rule	Statement	Purpose and justification
1	permit 192.168.0.0 0.0.255.255	Corporate & Guest Subnets Clearance: Identifies all end-user segments (192.168.0.0/16) across Manama, Riyadh/Dammam, and Kuwait (including Staff, Management, and Guest VLANs), permitting them to be dynamically translated to the public interface IP (200.1.1.1) for Internet browsing.
2	permit 10.1.0.0 0.0.255.255	Core Infrastructure & DMZ Clearance: Identifies core router loopbacks, inter-router transit links, and DMZ server segments (10.1.0.0/16), ensuring internal servers and control nodes can reach external WAN destinations via PAT when required.
implicit	deny ip any any	Unauthorized Source Block: Standard implicit deny rule that prevents any unlisted or spoofed private address space from consuming dynamic NAT translation slots on the edge router.

9.7 ACL tests (permitted and denied traffic)

Table 47: *ACL tests*

#	Test	Expected result
8.1	Ping `10.1.1.2` and `10.1.1.3` from any internal PC	**Fails** (intentional, rules 1–2)
8.2	Open the web page of `10.1.1.3` from an internal PC	Success
8.3	DNS query to `10.1.1.2` from a PC	Success
8.4	Guest laptop pings a Finance/HR/Compliance host	**Fails** (intentional)
8.5	Guest laptop reaches the DMZ web server and the Internet (`200.1.1.2`)	Success
8.6	`show access-lists` after the tests	Match counters increase on the deny and permit lines hit
8.7	SSH from VLAN 70 vs from another VLAN	Admin VLAN succeeds; others refused (Section 10)

```
M-R-edge#show access-lists
Extended IP access list EDGE_SECURITY_ACL
    10 deny icmp any host 10.1.1.2
    20 deny icmp any host 10.1.1.3 (4 match(es))
    30 deny ip 192.168.11.128 0.0.0.127 192.168.0.0 0.0.255.255
    40 deny ip 192.168.20.192 0.0.0.31 192.168.0.0 0.0.255.255
    50 deny ip 192.168.30.192 0.0.0.31 192.168.0.0 0.0.255.255
    60 permit ip 192.168.0.0 0.0.255.255 any
    70 permit ip 192.168.11.128 0.0.0.127 any
    80 permit ip 192.168.20.192 0.0.0.31 any
    90 permit ip 192.168.30.192 0.0.0.31 any
Standard IP access list NAT_ACL
    10 permit 192.168.0.0 0.0.255.255 (32 match(es))
    20 permit 10.1.0.0 0.0.255.255

M-R-edge#
```

```
M-R-dist1#show access-lists
Standard IP access list 10
    10 permit 192.168.11.96 0.0.0.31
Extended IP access list GUEST_IN
    10 permit udp any any eq bootps
    20 deny ip 192.168.11.128 0.0.0.127 192.168.0.0 0.0.255.255 (3 match(es))
    30 deny ip 192.168.11.128 0.0.0.127 10.1.0.0 0.0.255.255
    40 permit ip any any (1826 match(es))
```

Figure 14: *Evidence – `show access-lists` with match counters on M-R-edge / M-R-dist1*

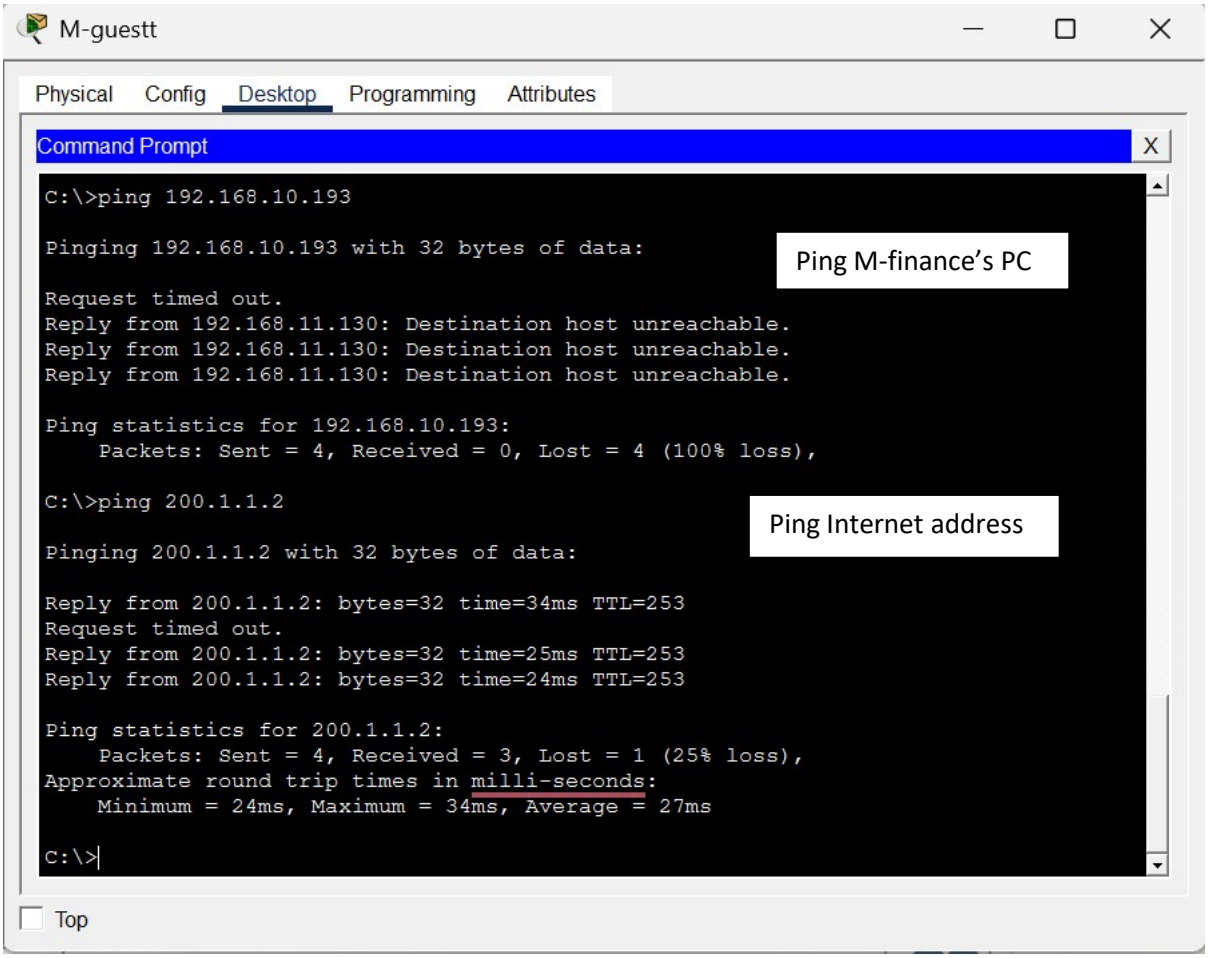


Figure 15: *Evidence – Failed ping from guest to Finance and successful guest browsing of the DMZ web server*

10. Secure Device Management (Requirement 9)

10.1 Theory: Telnet versus SSH, authentication and the management plane

The **management plane** is the set of protocols used to configure and monitor devices (SSH, Telnet, SNMP, HTTP). Anyone who controls it controls the network, so it must be authenticated, encrypted and restricted to authorised stations.

Table 48: *Telnet versus SSH*

Feature	Telnet	SSH v2
Encryption	None – credentials and commands visible to anyone capturing traffic	Strong encryption of the whole session
Authentication	Password only	Username/password or public key, host authentication
Integrity	None	Message authentication codes
Requirement	None	Hostname, domain name and RSA key pair
Decision	Disabled (`transport input ssh`)	**Selected** (`ip ssh version 2`)

10.2 Implementation

Every router and switch (except the simulated ISP) is configured with the same management baseline.

Table 50: *Elements of the management baseline*

Element	Command	Effect
Domain and keys	`ip domain-name antech.local`, RSA key pair	Required to generate SSH host keys
SSH version	`ip ssh version 2`	Only the more secure SSH v2 is accepted
Local account	`username admin secret AdminPass`	Individual login with a hashed secret
Privileged mode	`enable secret Antech`	Hashed password for privileged EXEC
Config protection	`service password-encryption`	Obscures remaining line passwords in the configuration
VTY lines (0–4 and 5–15)	`login local`, `transport input ssh`	All 16 virtual terminals require a local account and accept SSH only; Telnet is refused
Source restriction	`access-class 10 in`	Only the administrator subnet can open a session
Legal banner	`banner motd`	Warns that use is monitored; supports legal action
Idle timeout	`exec-timeout 5 0` (M-S-access1)	Closes forgotten sessions
Management address	`interface Vlan70` + `ip default-gateway`	Switches are managed in the Administrators VLAN (Appendix B)

10.3 Management tests

Table 52: *Management tests*

#	Test	Expected result
9.1	From an Administrators-VLAN PC: `ssh -l admin <device IP>`	Login prompt, then successful login
9.2	Same command from Finance, HR or a guest	Connection refused / times out (ACL 10)
9.3	`telnet <device IP>` from the administrator PC	Refused (`transport input ssh`)
9.4	`show ip ssh` and `show crypto key mypubkey rsa`	SSH version 2 enabled; key pair present

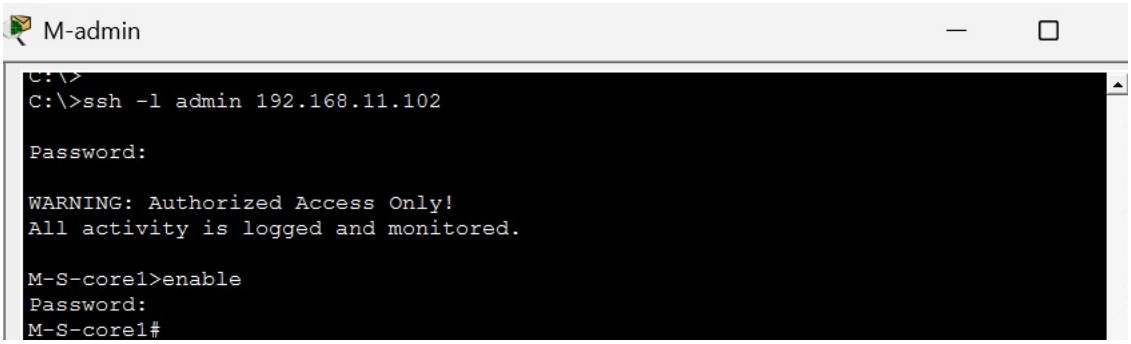


Figure 16: *Evidence – Successful SSH session from the Administrators VLAN*

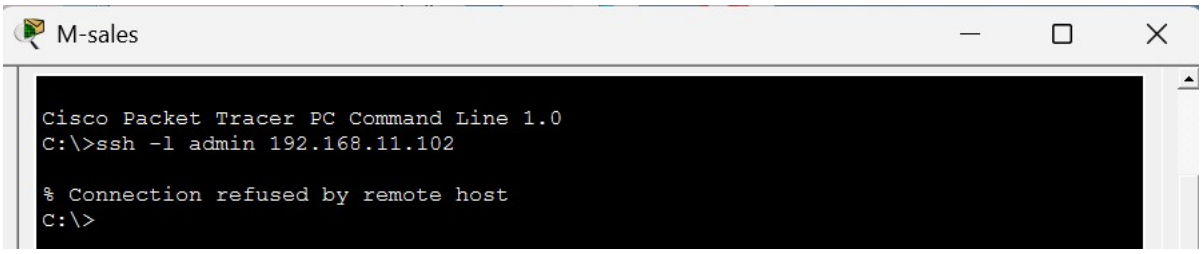


Figure 17: *Evidence – Refused SSH attempt from a non-administrator VLAN*

11. Switch Port Security (Requirement 10)

11.1 Theory: MAC learning and violation modes

A switch learns the source MAC address of every frame and stores it in its **CAM table**, which maps each MAC address to a port. An attacker can exploit this by plugging in an unauthorised device, spoofing a legitimate MAC address, or flooding the CAM table with thousands of fake addresses so that the switch behaves like a hub (**MAC flooding**). **Port security** limits which and how many MAC addresses may use a port.

11.2 Design decision and implementation

-Targeted Deployment: Port Security is enforced exclusively on end-user access ports (e.g., Staff, Finance, and Management VLANs).

-MAC Address Threshold & Persistence: Configured to allow a maximum of one MAC address (maximum 1), utilizing dynamic persistent learning (mac-address sticky) to automatically bind the connected host's identity to the interface running configuration.

-Violation Strategy: Set to shutdown mode. If an unauthorized device connects to an active access port, the switch interface immediately transitions into an err-disabled state, isolating the port until administrative intervention.

-Designated Exceptions: Trunk interfaces, EtherChannel bundles, and designated Wireless Access Point ports (such as Fa0/4 on Access Switch 1) are exempted from strict single-MAC limits to maintain multi-host transport and dynamic wireless client connectivity.

11.3 Port-security tests

Table 56: *Port-security tests*

#	Test	Expected result
10.1	`show port-security interface fa0/3` on M-S-access4	Enabled, maximum 1, sticky MAC learned, violation mode Shutdown
10.2	Disconnect the Finance PC and connect another PC to the same port	Port becomes `err-disabled`; link goes down
10.3	`show interfaces status err-disabled`; `show port-security`	Port listed with reason `psecure-violation`; violation count 1
10.4	Reconnect the original PC and recover with `shutdown` then `no shutdown`	Connectivity restored

```
M-S-access2#show port-security interface f0/3
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 1
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 1
Last Source Address:Vlan   : 0030.A36B.ADD8:20
Security Violation Count   : 0

M-S-access2#
```

```
M-S-access2#show port-security interface f0/3
Port Security              : Enabled
Port Status                : Secure-shutdown
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 1
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 1
Last Source Address:Vlan   : 000B.BE75.4356:20
Security Violation Count   : 1
```

Figure 18: *Evidence – `show port-security interface` and the err-disabled port after connecting an unauthorised PC*

12. First-Hop Redundancy – HSRP (Requirement 11)

12.1 Theory: the risk of a single default gateway

A host is configured with one default gateway address. If the router that owns that address fails, every host in the VLAN loses access to all other networks even though a second router is available. **First-hop redundancy protocols** solve this by giving two or more routers a shared **virtual IP address** (and virtual MAC address). Hosts use the virtual address as their gateway; one router is **active** (answers ARP for the virtual IP and forwards the traffic) and another is **standby** (monitors the active router and takes over when hellos stop).

Table 57: *FHRP comparison*

Feature	HSRP	VRRP	GLBP
Standard	Cisco proprietary (RFC 2281 informational)	Open standard (RFC 5798)	Cisco proprietary
Roles	Active / Standby	Master / Backup	AVG + several AVFs
Load balancing	Per VLAN/group only	Per group only	Per host automatically
Default timers	Hello 3 s, hold 10 s	Advertisement 1 s	Hello 3 s
Preemption	Optional (`preempt`)	On by default	Optional
Suitability	**Selected** – all HQ gateways are Cisco routers; widely documented; simple	Needed only for multi-vendor	More complex than needed

12.2 Design and implementation

`M-R-dist1` and `M-R-dist2` provide the gateway for all eight HQ VLANs. Each VLAN is its own HSRP group; the group number equals the VLAN ID, which makes the configuration self-documenting. The virtual IP is the first usable address of each subnet and is the gateway issued by DHCP. `M-R-dist1` is configured with priority 110 (higher than the default 100) and `preempt`, so it is the active router in normal operation and regains the role after it recovers. `M-R-dist1` is also the router attached to `M-S-core1`, which is the spanning-tree root, so Layer 2 and Layer 3 forwarding paths coincide and traffic does not take the inter-switch link unnecessarily.

Table 58: *HSRP groups (from the running-configs)*

VLAN / group	Virtual IP	M-R-dist1	M-R-dist2	Expected active
10	192.168.10.1	.2, priority 110, preempt	.3, priority 100 (default), preempt	M-R-dist1
20	192.168.10.129	.130, **priority not set (100)**, no preempt	.131, priority 100, preempt	**M-R-dist2** (tie broken by higher IP)
30	192.168.10.193	.194, priority 110, preempt	.195, priority 100, preempt	M-R-dist1
40	192.168.10.225	.226, priority 110, preempt	.227, priority 100, preempt	M-R-dist1
50	192.168.11.1	.2, priority 110, preempt	.3, priority 100, preempt	M-R-dist1
60	192.168.11.65	.66, priority 110, preempt	.67, priority 100, preempt	M-R-dist1
70	192.168.11.97	.98, priority 110, preempt	.99, priority 100, preempt	M-R-dist1
80	192.168.11.129	.130, priority 110, preempt	.131, priority 100, preempt	M-R-dist1

12.3 First-hop redundancy tests

Table 60: *HSRP tests*

#	Test	Expected result
11.1	`show standby brief` on both routers	dist1 Active / dist2 Standby for every group (VLAN 20 as documented)
11.2	Continuous ping from a PC to a remote host; shut `Gi0/2` of M-R-dist1	Few lost replies, then replies resume via M-R-dist2
11.3	`show standby brief` on M-R-dist2 during the failure	Active for all groups
11.4	Re-enable the interface on M-R-dist1	Because of `preempt`, dist1 becomes Active again

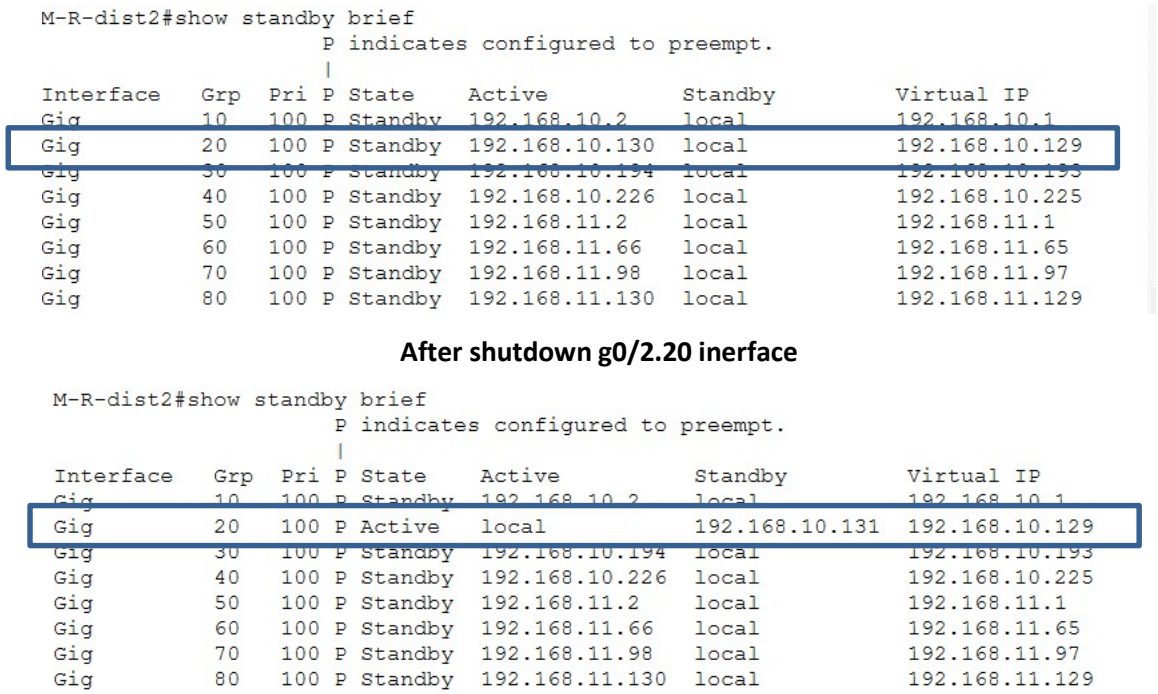


Figure 19: *Evidence – `show standby brief` before and during the failover test*

13. EtherChannel and Layer 2 Redundancy (Requirement 12)

13.1 Theory

EtherChannel (link aggregation, IEEE 802.1AX) bundles several physical links between two switches into one logical interface called a **port-channel**. Spanning tree treats the bundle as a single link, so all member links forward at the same time instead of all but one being blocked. The result is more bandwidth, load distribution (frames are distributed across members using a hash of addresses) and redundancy (if one member fails, the others continue).

Table 61: *Bundling protocols*

Mode	Protocol	Standard	Behaviour	Decision
on	None (static)	–	Forces the bundle with no negotiation; a mismatch can create loops	Rejected
`desirable` / `auto`	PAgP	Cisco	Negotiates; Cisco switches only	Rejected
`active` / `passive`	LACP	IEEE 802.3ad / 802.1AX	Negotiates, detects misconfigured members, multi-vendor	**Selected** (`active` on both ends)

13.2 Implementation

Two member links (`Fa0/3` and `Fa0/4`) are bundled into `Port-channel1` between the two core switches at HQ and between the two switches at each branch. `channel-group 1 mode active` is used on both ends so that LACP negotiation verifies that both sides agree before traffic is forwarded. The two Fast Ethernet members provide a logical 200 Mbps link.

```
interface Port-channel1
 switchport trunk allowed vlan 10,20,30,40,50,60,70,80
 switchport mode trunk
interface FastEthernet0/3
 switchport trunk allowed vlan 10,20,30,40,50,60,70,80
 switchport mode trunk
 channel-group 1 mode active
interface FastEthernet0/4
 switchport trunk allowed vlan 10,20,30,40,50,60,70,80
 switchport mode trunk
 channel-group 1 mode active
```

Table 62: *EtherChannel bundles*

Site	Switches	Members	Mode	Trunk VLANs
Manama	M-S-core1 ↔ M-S-core2	Fa0/3, Fa0/4	LACP active	10–80
Dubai	D-switch1 ↔ D-switch2	Fa0/3, Fa0/4	LACP active	110–140
Kuwait	K-switch1 ↔ K-switch2	Fa0/3, Fa0/4	LACP active	210- 240

Together with Rapid PVST+ (Section 3.5) and the dual uplinks of every HQ access switch, this gives a loop-free Layer 2 design in which any single cable or switch-port failure is tolerated. The access switches are dual-homed to the two cores; spanning tree blocks one of the two uplinks and unblocks it on failure, while the core-to-core link uses EtherChannel so it never becomes a bottleneck.

13.3 EtherChannel tests

Table 63: *EtherChannel tests*

#	Test	Expected result
12.1	`show etherchannel summary`	Po1 (SU), Fa0/3(P), Fa0/4(P), protocol LACP
12.2	`show interfaces trunk`	Port-channel1 listed as the trunk, not the individual members
12.3	Continuous ping across the bundle; disconnect one member	No (or one) lost reply; Po1 stays up with one member
12.4	Reconnect the member	Member rejoins the bundle automatically

```
M-S-core1#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------

1      Po1(SU)           LACP    Fa0/3(P) Fa0/4(P)
M-S-core1#
```

Figure 20: *Evidence – `show etherchannel summary`*

14. Network Services and Servers (Requirement 13)

14.1 Theory: common enterprise services

Table 64: *Enterprise network services*

Service	Protocol / port	Role in this network
DHCP	UDP 67/68	Automatically assigns address, mask, gateway and DNS to hosts, removing manual errors
DNS	UDP/TCP 53	Translates names to addresses; server `10.1.1.2` in the DMZ is given to every client
Web (HTTP/HTTPS)	TCP 80/443	Public web site hosted on `10.1.1.3`
Email (SMTP/POP3)	TCP 25 / 110	Mail server on `10.1.1.3` (combined Email/Web server)
Management services	SSH, SNMP, syslog, NTP	SSH is deployed; SNMP/syslog/NTP are discussed in Section 15

14.2 DHCP implementation

DHCP is configured **on the distribution routers** (`M-R-dist1` for HQ, `D-R-dist` for Dubai, `K-R-dist` for Kuwait). Hosting the service on the same router that is the gateway means no DHCP relay (`ip helper-address`) is needed and the number of devices to maintain is minimal. Every pool gives the HSRP virtual IP (or the branch gateway) as the default router and the DMZ DNS server `10.1.1.2`. The first addresses of each subnet are excluded from the pool so they can be used for routers, HSRP, switches and static devices.

Table 65: *Manama DHCP pools (M-R-dist1)*

Pool	Network	Default router	DNS	Excluded addresses	Leasable addresses
VLAN10_POOL	192.168.10.0/25	192.168.10.1	10.1.1.2	.1 – .3	123
VLAN20_POOL	192.168.10.128/26	192.168.10.129	10.1.1.2	.129 – .135	55
VLAN30_POOL	192.168.10.192/27	192.168.10.193	10.1.1.2	.193 – .199	23
VLAN40_POOL	192.168.10.224/27	192.168.10.225	10.1.1.2	.225 – .231	23
VLAN50_POOL	192.168.11.0/26	192.168.11.1	10.1.1.2	.1 – .10	52
VLAN60_POOL	192.168.11.64/27	192.168.11.65	10.1.1.2	.65 – .71	23
VLAN70_POOL	192.168.11.96/27	192.168.11.97	10.1.1.2	.97 – .106	20
VLAN80_POOL	192.168.11.128/25	192.168.11.129	10.1.1.2	.129 – .140	114

Table 66: *Branch DHCP pools*

Pool (router)	Network	Default router	Excluded	Leasable
D_POOL_VLAN110 (D-R-dist)	192.168.20.0/25	192.168.20.1	.1 – .10	116
D_POOL_VLAN120	192.168.20.128/26	192.168.20.129	.129 – .138	52
D_POOL_VLAN130	192.168.20.192/27	192.168.20.193	.193 – .200	22
D_POOL_VLAN140	192.168.20.224/27	192.168.20.225	.225 – .230	24
K_POOL_VLAN210 (K-R-dist)	192.168.30.0/25	192.168.30.1	.1 – .10	116
K_POOL_VLAN220	192.168.30.128/26	192.168.30.129	.129 – .138	52
K_POOL_VLAN230	192.168.30.192/27	192.168.30.193	.193 – .200	22
K_POOL_VLAN240	192.168.30.224/27	192.168.30.225	.225 – .230	24

```
ip dhcp excluded-address 192.168.10.129 192.168.10.135
ip dhcp pool VLAN20_POOL
 network 192.168.10.128 255.255.255.192
 default-router 192.168.10.129
 dns-server 10.1.1.2
```

14.3 DNS, Web and Email servers in the DMZ

Table 67: *DMZ servers*

Server	Address	Switch port	Services	Gateway
DNS server	10.1.1.2/28	M-S-DMZ `Fa0/1` or `Fa0/2` (VLAN 15)	DNS	10.1.1.1 (M-R-edge `Gi0/1.15`)
Email/Web server	10.1.1.3/28	M-S-DMZ `Fa0/2` or `Fa0/1` (VLAN 15)	HTTP, HTTPS, SMTP, POP3	10.1.1.1

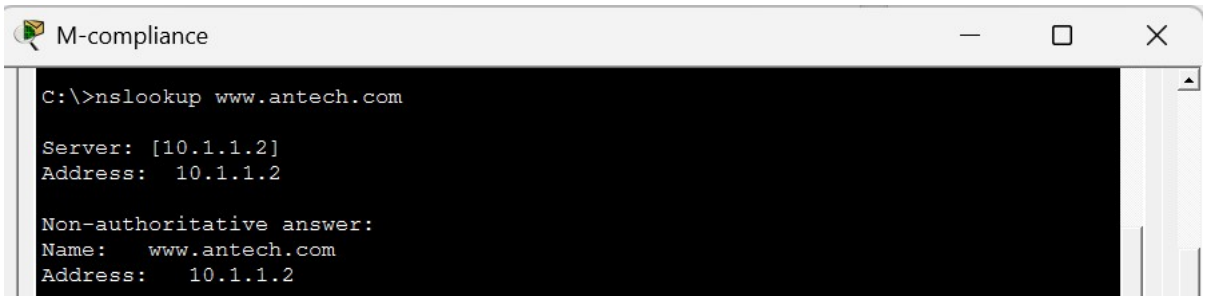
Why a DMZ? Servers that must be reachable from outside are placed in their own subnet (`10.1.1.0/28`, VLAN 15) behind a dedicated router interface. Even if a server is compromised, the attacker is still on a segment that is separate from the internal departments, and the router can filter all traffic entering or leaving it. The DMZ switch ports are PortFast access ports in VLAN 15, and all other ports are shut down.

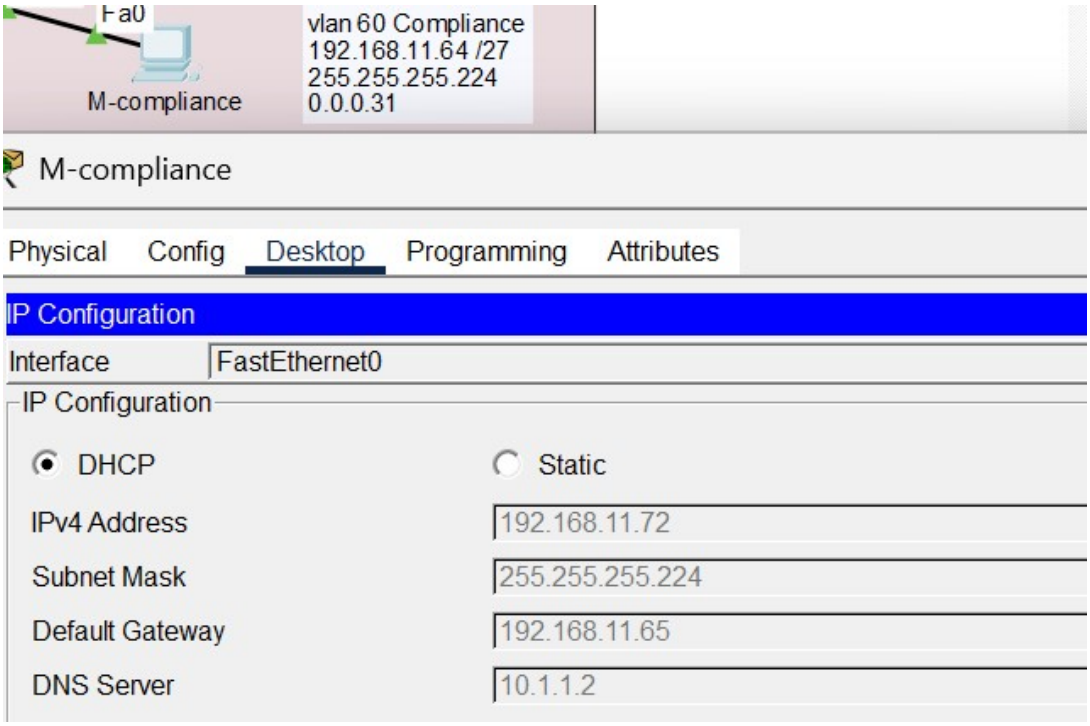
Table 68: *Controls that protect the services*

Control	Effect
Separate VLAN 15 / subnet behind M-R-edge	Servers isolated from user VLANs
`EDGE_SECURITY_ACL` (ICMP deny, Section 9)	Servers cannot be pinged or mapped by ping sweeps; services still available
NAT / static translation (Section 8)	Only chosen ports are exposed on the public address
DMZ switch management restricted by ACL 10 and SSH	Only administrators can configure the switch
Unused DMZ switch ports shut down	No unauthorised device can join the DMZ

Table 69: *Service tests*

#	Test	Expected result
13.1	`ipconfig /all` on a PC set to DHCP	Address in the VLAN range outside the excluded block, gateway = HSRP VIP, DNS = 10.1.1.2
13.2	`nslookup <server name>` from an internal PC	Resolves to the DMZ address
13.3	Browse `http://<name>` from HQ, Dubai and Kuwait PCs	Web page opens
13.4	Send and receive an e-mail between two clients	Delivered
13.5	Ping `10.1.1.2` / `10.1.1.3`	Fails (intentional ICMP rule)





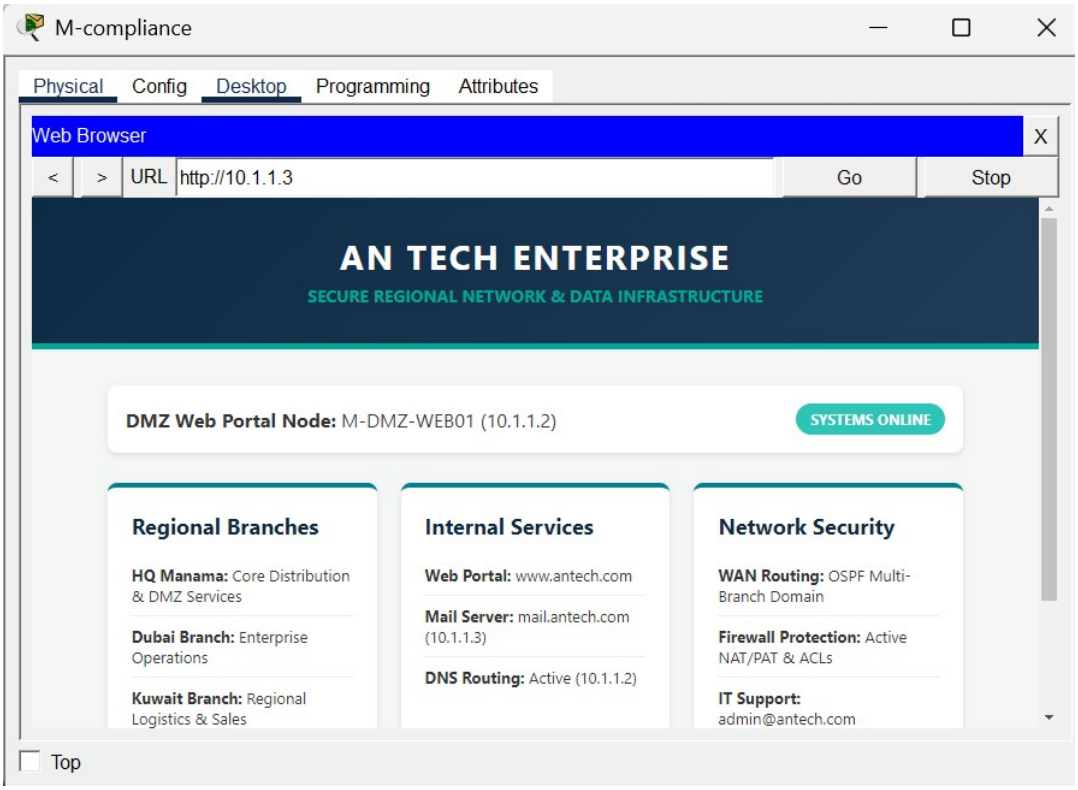


Figure 21: *Evidence – DHCP lease on a client and DNS/web test*

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